

The role of online banking in promoting financial inclusion in Bangladesh.”



A thesis

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CERTIFICATION

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.....

Supervisor

DEDICATION

This work is dedicated to my parents.

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Abstract

Financial inclusion has been identified as an essential pillar of sustainable economic growth as it enhances access to formal financial services and facilitates economic inclusion for all. Bangladesh has seen a contentious growth of online banking and digital financial services, altering the landscape of the financial sector, but empirical research on the long-run financial inclusion impact of online banking and digital financial services is scarce. This study aims to explore how online banking can be used to foster financial inclusion in the Bangladeshi context using the impact of active mobile financial services accounts, automated teller machines (ATMs) and commercial bank branches on deposit account growth as an indicator of financial inclusion. Bangladesh Bank data were used for this purpose and were collected for the past 10 years (2014-2023). The evolution of the indicators of financial inclusion was analyzed using descriptive statistics and trend analysis, while the stationarity properties of the variables were examined using the Augmented Dickey–Fuller (ADF), Phillips–Perron (PP) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests. ARDL model and associated Error Correction Model (ECM) were used to examine the long-run equilibrium and short run dynamics between the variables. In addition, an Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model was used to forecast future deposit account growth through 2030. The descriptive analysis showed significant growth in the indicators on financial inclusion during the study period, as the number of deposit accounts went up from 70.01 million to 149.48 million, the number of mobile financial service accounts which were active increased from 9.50 million to 83.69 million, and the number of ATMs across the country increased from 6259 to 17471. The ARDL results reveal that financial inclusion of current years' deposit accounts is significantly influenced with its previous years' level (coefficient is 0.7454 and P value is less than 0.01) which proves the high persistence of financial inclusion. This indicates that the estimated coefficients of commercial bank branches and active mobile estimated coefficients have positive and statistically significant short-run impacts on the growth of deposits, with AR coef. of 3.0287 and 0.1790 respectively and with p-value < 0.05. This implies that physical banking infrastructure plays a role in financial inclusion after some time lag, whereas the ARs of mobile financial services display a positive effect on the growth of deposits after some time lag. The Bounds test ($F = 4.1476$) supports a long-run relationship at a 10% significance level, while the negative and statistically significant error correction term (-0.2832 , $p < 0.01$) suggests that there is a mean rate of error correction of approximately

28.3% given a one-year lag. Additionally, the ARIMAX forecasting model suggests that the number of financial inclusion accounts will show continued growth throughout the forecast period, from 155.63 million in 2024 to around 200.62 million by 2030. The results are similar to the previous studies which have shown that digital financial services are a vital contributor to financial inclusion in developing economies and that traditional financial services will still play a complementary role. The study points out that online banking, in particular, mobile financial services have emerged as a key force in enhancing financial inclusion in Bangladesh. This brings together the econometric modelling and forecasting tools, using official Bangladesh Bank data to provide empirical evidence which can be used to craft policies and programs to scale up provision of inclusive and sustainable digital financial services for policy makers and financial institutions to grow the country's digital financial services to 2030 and beyond.

Keywords: Financial Inclusion, Online Banking, Digital Financial Services, Mobile Financial Services, ARDL, ARIMAX, Forecasting, Bangladesh.

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Chapter One

Introduction

1.1 Background

Financial Inclusion in the Global Context

Financial inclusion has emerged as one of the most significant development priorities for governments, international organizations, and academic researchers over the past two decades. In 2005, the United Nations declared it the **International Year of Microcredit**, highlighting the significance of the availability of financial services in the fight against poverty and inclusive economic growth. In general, financial inclusion is the availability of useful, affordable, and appropriate formal financial services and products, especially savings, payments, credit, and insurance, provided in a responsible and sustainable way to individuals and businesses (*The World Bank Annual Report 2014, 2014*). The (*Annual Report 2022, n.d.*) also notes that this should go to those who have previously been excluded from the formal financial system. The importance of financial inclusion is that it enables individuals and households to deal with financial risks, smooth consumption, save and invest in productive activities. At the macro level, financial systems that are inclusive have been shown to have a positive impact on economic growth, through the process of mobilizing savings, better capital allocation, and better financial stability (**Levine, 2005**).

However, there has been significant progress, and financial exclusion is still a challenge everywhere. As per the Global Findex Database, about 1.4 billion adults were not banked in 2021, most of whom are from low and middle-income countries (**Demirgüç-Kunt et al., 2022**). Financial exclusion is more prominent in rural areas, among women, low income families and informal sector jobs. Structural barriers such as low income, lack of identification, geographic isolation, and limited financial literacy continue to impede access to formal financial services.

The importance of financial inclusion is in helping individuals and families to protect themselves against financial risks, hedge consumption, save, and invest in productive activities. Empirical studies have consistently demonstrated that access to formal finance improves household welfare by facilitating investment in education, healthcare, and small, scale entrepreneurship (**Beck, Demirguc-Kunt, et al., 2007**). At the macroeconomic level, inclusive

financial systems contribute to economic growth by mobilizing savings, improving capital allocation, and enhancing financial stability **(Levine, 2005a)**.

Despite considerable progress, financial exclusion remains a persistent challenge worldwide. According to the Global Findex Database, approximately 1.4 billion adults globally remained unbanked as of 2021, with the majority residing in low, and middle, income countries **(Demirgüç-Kunt et al., 2022)**. Financial exclusion is a specific problem for women, informal sector workers, rural people and low income groups. The ability of people to access formal financial services is still constrained by the structural barriers of low income, lack of identification, geographic isolation and limited financial literacy.

The traditional banking system of brick and mortar branches has been unable to meet these challenges well, particularly in the developing economies. An expansion into rural areas is not an attractive option for banks, as they would face high costs in operation, low density and low number of transactions. As a result, other financial service delivery methods have become more prominent, especially those that utilize technology innovations to cut costs and increase reach.

Financial Inclusion in Developing Economies

The financial inclusion challenge is greater in developing economies because of structural, institutional and socio-economic constraints. Many low income countries have a large informal financial sector, consisting of moneylenders, rotating savings groups, and informal credit networks. These mechanisms offer short-term remedies, but they are also expensive and may not offer much protection to consumers.

From the demand side, low and irregular income, lack of financial literacy, and mistrust in formal financial institutions discourage participation in formal banking systems. From the supply side, financial institutions face high fixed costs, regulatory constraints, and operational risks in extending services to underserved areas **(Beck & De La Torre, 2007)**. Distance to bank branches significantly increases transaction costs, reducing the effective demand for financial services among rural households **(Allen et al., 2016a)**. The use of digital financial services is increasingly being seen as a viable solution to these challenges. Financial service providers have been able to serve populations not previously served at lower marginal costs, thanks to the development of mobile technology, electronic payment systems, and digital banking platforms. Cross, country evidence suggests that digital finance can significantly

improve access to financial services, particularly in regions with limited physical banking infrastructure (Ozili, 2018).

Overview of the Banking Sector and Financial Inclusion in Bangladesh

Bangladesh has experienced sustained economic growth over the past decade, emerging as one of the fastest, growing economies in South Asia. The country recorded GDP growth of 8.2 percent in 2019, driven by industrial expansion, export growth, and remittance inflows (Bhuiyan & Badran, 2020). The banking sector plays a central role in financial intermediation and economic development, accounting for the majority of financial assets.

Key barriers to financial inclusion in Bangladesh include minimum balance requirements, lack of Despite this progress, financial inclusion in Bangladesh remains limited. According to more recent data from the **Global Findex Database (2021)**, about 53% of Bangladeshi individuals have access to a bank or mobile money account, which is a significant improvement from previous years. According to the Global Findex Database, approximately 59 percent of adults aged 15 and above did not have an account with a formal financial institution in 2017, although this represented an improvement from 71 percent in 2014 (A. Demirguc-Kunt et al., 2018).

Awareness and limited financial literacy and inadequate physical and technological infrastructure are higher in rural areas, among women, and among low-income households, with financial exclusion being disproportionately higher among these groups. They are especially severe in rural and remote areas where formal banking services are less widely available. With COVID, 19, the pandemic has continued to drive the growth of online and digital financial services, underscoring its role in providing continued financial access.

Conventional Channels of Financial Inclusion in Bangladesh

Branch Banking

It is in general that banking system in Bangladesh has been delivering banking services through the branch banking system. To promote rural financial inclusion, Bangladesh Bank issued a circular in 2011 mandating a 1:1 ratio in establishing new bank branches in urban and rural areas. As of 2019, the number of commercial bank branches per 100,000 adults stood at approximately 9, with a total of 10,588 branches operating nationwide by fiscal year 2020 (IMF Financial Access Survey).

Although branch expansion has contributed to incremental improvements in access, it remains a costly and slow process. High fixed and operational costs limit the scalability of branch, based strategies, making them insufficient for achieving universal financial inclusion (**Beck, Demirgüç, Kunt, et al., 2007**).

Agent Banking

Agent banking has emerged as a complementary channel to extend financial services to underserved areas. As of June 2020, 23 commercial banks were operating agent banking services through 8,764 agents, primarily serving rural communities (**Bangladesh Bank, 2020**). While agent banking reduces operational costs and improves outreach, its effectiveness depends on regulatory oversight, agent density, and customer trust. Agent banking outlets offer individualized services without the need for physical bank branches

Digital Channels of Financial Inclusion

Automated Teller Machines (ATMs)

The advent of automated teller machines (ATM) has been instrumental to the development of electronic banking in Bangladesh since 1993. ATMs allow customers to conduct simple transactions without setting foot inside a branch, saving time and transactional costs. By 2019, for every 100,000 adults there were now 9.39 ATMs compared to the number of bank branches outpacing them according to **IMF** data. Empirical studies suggest that ATM expansion enhances financial inclusion by improving convenience and reducing dependence on branch, based services, particularly in semi, urban and rural areas (**Rahman et al., 2022a**)

Mobile Financial Services (MFS).

Mobile financial services represent one of the most transformative innovations in the financial sector. In Bangladesh, MFS was formally launched in March 2011 under a bank, led regulatory framework. Platforms such as **bKash, Rocket, Nagad, and Upay** have become household names, allowing millions to save, transfer money, and make payments. Mobile banking enables users to conduct transactions remotely, making it particularly suitable for geographically dispersed populations.

Despite rapid growth, adoption remains limited. Data from the Bangladesh Telecommunication Regulatory Commission indicate that only 14.3 percent of mobile subscribers were registered

MFS users in 2014. Similar adoption constraints have been observed in other developing economies due to trust issues, limited digital literacy, and infrastructural gaps (**Rahman et al., 2022**).

FinTech, Digitalization, and Financial Inclusion

The rapid advancement of financial technology (FinTech) has transformed global financial systems. FinTech innovations-including mobile wallets, digital payments, online lending platforms, and data, driven financial services-have expanded access to finance and reduced exclusion (**Arner et al., 2020**). Digital finance is increasingly recognized as a key enabler of inclusive economic growth and a critical contributor to the United Nations Sustainable Development Goals.

In Bangladesh, Bangladesh Bank has introduced several initiatives to promote digital finance, including MFS guidelines, agent banking regulations, and digital payment infrastructure. This expansion aligns with the "Digital Bangladesh" vision, aiming to reduce inequality and empower marginalized groups. However, empirical evidence assessing the long, term impact of these initiatives remains limited.

1.2 Context

Financial inclusion has emerged as a key global development agenda as access to formal financial services is critical for countries' economic growth, poverty alleviation and social wellbeing. The World Bank and other international bodies have recognized the importance of financial inclusion to help people and businesses save money safely, obtain credit and access payment services in an efficient way, and manage financial risks. Despite the considerable gains achieved over the past few years, many people in developing countries are still not included in the formal financial system.

Over the past decade, Bangladesh has made tremendous progress in its economic development and digitalization. The banking industry has grown significantly in all of the traditional ways of banking, including the Automated Teller Machines (ATMs), Internet banking, agent banking and mobile financial services. Formal financial services access is not equal, however, with low, income groups and rural and remote populations less likely to access. Historically, financial inclusion has been limited in Bangladesh because of geographical accessibility to bank branches and lack of financial skills and knowledge, as well as limited banking infrastructure.

To overcome these challenges, Bangladesh Bank has taken a few initiatives to encourage digital financial services. Mobile banking, Internet banking, agent banking and ATM networks have changed the way financial services are delivered and are growing at an increasing pace. In addition, mobile financial services in particular have cut costs in transactions and made it possible for individuals to carry out financial transactions without having to visit a bank branch. This has led to digital banking becoming an important tool in the arms of banks to increase financial access and financial non, exclusion.

As more and more financial institutions are moving to digital platforms, an important question is emerging on how digital technologies contribute to the financial inclusion. The study has also investigated how financial inclusion in Bangladesh is affected by the digital banking infrastructure based on recent macro, level data, and while literature exists on mobile financial services, studies on internet banking adoption and digital financial services are found separately, this study aimed to examine the financial inclusion and impact of digital banking infrastructure. So, in this regard, this paper is focused on the role of online banking and digital financial services in financial inclusion in Bangladesh.

Annual data from 2014 to 2023 is used with Deposit Accounts of Commercial Banks as proxy for financial inclusion and Active Mobile Money Accounts, Automated Teller Machines (ATMs), and Commercial Bank Branches as key determinants of financial access. The study uses the Autoregressive Distributed Lag (ARDL) model to examine the short run and long run relationships. Moreover, the ARIMAX model can be used to predict the future trend of financial inclusion up to 2028. The findings will give valuable insights to the policymakers, financial institutions, and regulators about the effectiveness of the digital banking initiatives in enhancing financial inclusion in Bangladesh.

Problem Statement

Financial inclusion is a key to sustainable economic development, poverty alleviation and enhanced living standards. Although the banking sector has advanced, there remains a large segment of the population in Bangladesh who have problems accessing formal financial services. The traditional banking sector, especially branch banking is unable to properly cater for the rural and remote regions because of the high operation costs, lack of infrastructure, and geographical limitations.

Bangladesh experienced a fast pace of digital financial services development, such as mobile banking, internet banking and ATM network, in recent years. These technologies have opened new possibilities to deliver financial services in a more efficient and cost, effective way. However, the contribution of these digital banking channels to financial inclusion is not so well studied with recent macroeconomic data.

While the two themes have been explored through separate research on mobile banking adoption and digital finance and on financial inclusion, there is little empirical evidence on the effect of the digital banking infrastructure on financial inclusion in Bangladesh. Thus, it is important to study the effect of mobile financial services, ATM networks and banking infrastructure on enhancing financial inclusion. To fill this gap this study aims to explore the relationship between digital banking indicators and Bangladesh's financial inclusion.

1.3 Purposes

This study aims to explore the role of online banking and digital financial services to enhance financial inclusion in Bangladesh. This study examines the relationship between the growth of digital banking infrastructure and access to formal financial services, proxied by having a deposit account in a commercial bank.

In particular, the study examines the effects of having an active mobile money account, the use of automated teller machines (ATM) and commercial bank branches on financial inclusion. Moreover, the study examines the short run and long run relationships between these variables with the help of Autoregressive Distributed Lag (ARDL) model. The study also uses the ARIMAX forecasting model to forecast future trends in financial inclusion in terms of deposit accounts, in order to evaluate future prospects of financial inclusion.

1.4 Research Objectives

Main Objective:

To explore the contribution of online banking towards financial inclusion in the context of Bangladesh.

Specific Objectives

1. To examine the financial inclusion and digital banking indicators in Bangladesh from 2014-2023.
2. To explore the effect of active mobile money accounts on financial inclusion.
3. To measure the impact of ATM networks on financial inclusion.
4. To assess and analyse the contribution of commercial bank branches to financial inclusion.
5. To examine the short run and long run relationships between digital banking indicators and financial inclusion based on ARDL model.
6. To predict financial inclusion based on ARIMAX model.

1.5 Work Flow

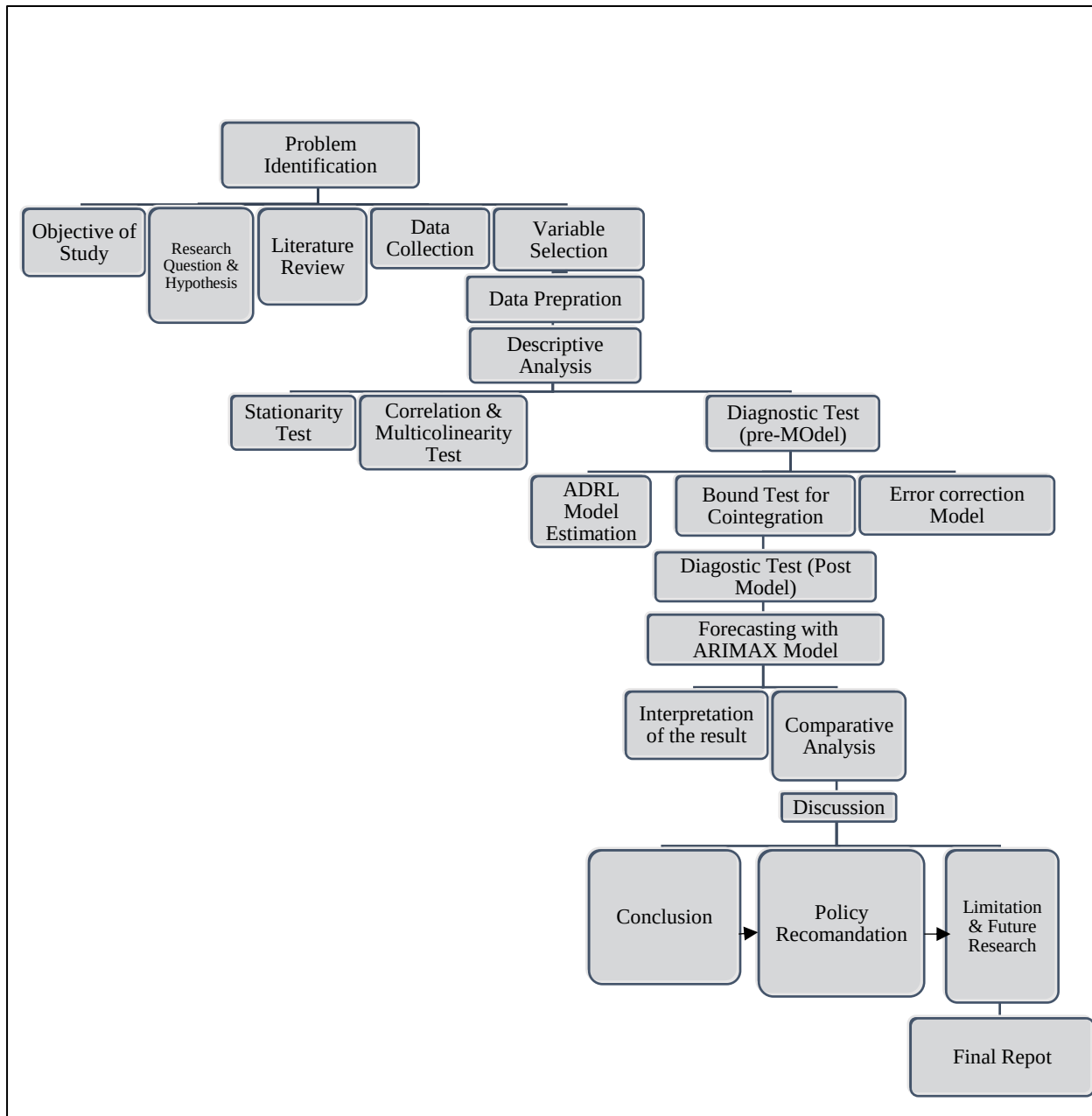


Figure 1: Work Flow

Chapter Two

Literature Review

2.1 Conceptualising Financial Inclusion

Financial inclusion is a multidimensional concept that extends beyond mere ownership of a bank account. **The World Bank (2014)** defines financial inclusion as access to useful and affordable financial products and services that meet individuals' needs, transactions, payments, savings, credit, and insurance, delivered responsibly and sustainably. This definition highlights three critical dimensions: access, usage, and quality of financial services. **(Sarma, 2012)** further defines it as a process that ensures the ease of access, availability, and usage of the formal financial system for all members of an economy.

From an academic perspective, access refers to the availability of financial services, often measured by indicators such as the number of bank branches, ATMs, or financial agents per population. Usage captures the actual adoption and frequency of use of financial services, typically reflected in the number of deposit accounts, active users, or transaction volumes. Quality relates to affordability, convenience, and consumer protection, though it is harder to quantify at the macro level.

Several scholars argue that financial inclusion is not an end in itself but a means to achieve broader economic and social objectives. Beck, **(Beck, Demirgüç, Kunt, et al., 2007b)** demonstrate that inclusive financial systems reduce income inequality and promote pro, poor growth by enabling access to productive investment opportunities. Similarly, **(Levine, 2005a)** emphasizes that financial development improves resource allocation efficiency, thereby accelerating long, term economic growth. Ultimately, the goal is to link marginalized and disenfranchised groups to official institutions to lessen social vulnerability and poverty.

In developing economies, financial inclusion is often constrained by structural barriers such as geographic isolation, low income, and weak institutional frameworks. Rural populations are particularly disadvantaged due to limited physical banking infrastructure and high transaction costs. As a result, conventional banking models struggle to achieve universal inclusion without significant cost escalation.

2.2 Supply, Side and Demand, Side Barriers to Financial Inclusion

The literature identifies financial exclusion as the outcome of both supply, side and demand, side constraints. On the demand side, low and irregular income, lack of financial literacy, mistrust in formal institutions, and cultural factors discourage individuals from engaging with formal financial systems (Allen et al., 2016b). According to (A. & K. Demirguc, Kunt, 2017), an individual's inability to access these systems hampers their capacity to save, invest, and build resilience against financial shocks. For rural households, opportunity costs associated with travel time and distance to bank branches further reduce effective demand.

On the supply side, banks face high fixed costs in establishing and maintaining branches in sparsely populated rural areas. (Beck et al., 2006) argue that traditional banking technologies are ill-suited for serving low-income clients due to high per-transaction costs and information asymmetries. Regulatory compliance costs and risk management requirements further limit outreach.

Empirical evidence from developing countries suggests that branch expansion alone yields only marginal improvements in financial inclusion and is financially unsustainable in the long run. This has prompted policymakers and researchers to explore alternative delivery channels that leverage technology to reduce costs and expand outreach.

2.3 Role of Conventional Banking Infrastructure

2.3.1 Bank Branches

Bank branches remain a fundamental component of financial systems, serving as access points for deposits, credit, and payment services. Studies using cross-country data find a positive association between branch density and financial inclusion, particularly in early stages of financial development (A. Demirguc, Kunt et al., 2018).

However, in rural contexts, branch-based models exhibit diminishing returns. (Allen et al., 2016a) note that beyond a certain threshold, additional branches contribute little to inclusion while significantly increasing operational costs. This trade-off is especially relevant in Bangladesh, where rural branch expansion is regulated but costly.

2.3.2 Automated Teller Machines (ATMs)

ATMs represent an intermediate technological solution that reduces reliance on branch banking. Empirical studies show that ATM density is positively associated with account usage and transaction frequency (**Rahman et al., 2022b**). While ATMs improve convenience, their effectiveness can be limited by technical challenges, transaction security concerns, and geographic placement. ATMs improve convenience and reduce transaction costs, but their effectiveness in rural inclusion is limited by electricity access, maintenance costs, and geographic placement.

Critically, ATMs still require physical presence and thus do not fully eliminate distance barriers faced by remote populations.

2.4 Digital Financial Services and Financial Inclusion

2.4.1 Mobile Financial Services (MFS)

Mobile financial services have emerged as one of the most significant innovations in promoting financial inclusion in developing economies. Mobile banking enables users to perform financial transactions remotely using mobile devices, thereby drastically reducing transaction costs and geographic barriers.

Donovan (2012) argues that mobile money platforms are particularly effective in reaching unbanked populations because they leverage existing mobile phone penetration. (**Ozili, 2018**) provides empirical evidence that digital finance significantly enhances financial inclusion by increasing account ownership and transaction usage.

In Bangladesh, MFS operates under a bank, led regulatory framework, ensuring systemic stability while promoting innovation. Studies focusing on Bangladesh find rapid growth in mobile money transactions and registered accounts, indicating strong potential for inclusion (**Ahmed & Enamul Hasan, 2021**). However, adoption remains uneven, particularly in rural areas, due to digital literacy gaps and trust issues. (**Gomber et al., 2017**) highlight that these services transform the financial ecosystem by enabling cashless transactions. In Bangladesh, MFS operates under a bank, led regulatory framework, ensuring systemic stability. Platforms like bKash, Nagad, and Rocket have become central to the digital transaction space, with some studies suggesting that DFI also improves financial stability and boosts government tax revenues by formalizing transactions (**Manyika, J., Singer, M. et al. (2016)**).

2.4.2 FinTech and Digitalisation

Beyond mobile money, broader FinTech innovations, including digital payments, online banking, and agent, based platforms, are transforming financial service delivery. Arner et al. (2020) emphasise that FinTech reduces marginal costs of service delivery, enabling scalable inclusion. Digital finance also aligns with the United Nations Sustainable Development Goals by facilitating inclusive economic participation.

Despite global evidence, the empirical literature on FinTech, driven inclusion in Bangladesh remains limited and fragmented, often focusing on adoption behaviour rather than macro, level outcomes. Recent research suggests that while digital financial services have spectacularly expanded inclusion, stubborn roadblocks like digital illiteracy and cyber fraud persist.

2.5 Empirical Evidence from Bangladesh

Existing studies on Bangladesh primarily adopt cross, sectional survey methods to examine adoption and usage of digital financial services. **(Hassan, K.A., Hasan, M.K., Chowdhury, M.G., et al. (2010) Aspects of Infection in Intensive Care Unit-Prevention and Control. Mymensingh Medical Journal, 19, 474, n.d.)** analyze customer satisfaction with internet banking, while **(Rahman, M. (2009) Contributions of Exports, FDI and Expatriates' Remittances to Real GDP of Bangladesh, India, Pakistan and Sri Lanka. Southwestern Economic Review, 36, 141, n.d.)** discusses policy implications of e, banking. Bilkis and Khan (2016) examine challenges in mobile financial services, highlighting regulatory and trust, related issues.

More recent studies explore specific technologies such as ATMs **(Rahman et al., 2022b)** or regulatory sandbox initiatives **(Azim, 2019)**. However, these studies often lack a dynamic perspective and do not analyze long, term trends using macro, level data.

Notably, very few studies employ time, series data to examine how digital banking infrastructure influences financial inclusion over time, nor do they attempt to forecast future inclusion trajectories.

2.6 Research Gap

Based on the reviewed literature, several critical gaps are identified:

1. Methodological gap: Most studies on Bangladesh rely on cross, sectional survey data, limiting their ability to capture dynamic relationships and long, term trends.
2. Data gap: Limited use of macro, level indicators such as ATM density, mobile money transactions, and active accounts.
3. Forecasting gap: Virtually no study attempts to forecast financial inclusion using time, series econometric models.
4. Integration gap: Existing literature treats conventional banking and digital channels separately rather than within a unified analytical framework.
5. This study addresses these gaps by employing time, series data to examine and forecast the role of online and digital banking infrastructure in promoting financial inclusion in Bangladesh.

2.7 Hypotheses Development

Based on theory and empirical evidence, the following hypotheses are formulated:

H₁: An increase in the number of commercial bank branches has a positive effect on financial inclusion in Bangladesh.

H₂: An increase in the number of automated teller machines (ATMs) positively influences financial inclusion.

H₃: Growth in mobile financial services, measured by active mobile money accounts and transaction value, significantly enhances financial inclusion.

H₄: Digital financial services have a stronger long, run impact on financial inclusion than conventional banking infrastructure.

H₅: Financial inclusion indicators exhibit a stable time, series pattern that can be forecasted reliably using appropriate econometric models.

Chapter Three

Methodology

3.1 Introduction

In this chapter, the methodology used to explore the role of Online Banking for achieving Financial inclusion in Bangladesh is discussed. It defines the type of research design, data sources, variables, analytical framework and statistical methods used in the study. Annual time, series data is used for the analysis for the previous 5 years (2014, 2023). The study takes descriptive statistics, trend analysis, unit root test, the autoregressive distributed lag (ARDL) model, and autoregressive integrated moving average with exogenous variables (ARIMAX) model in order to look at the dynamic relationship and the future trend of financial inclusion.

3.2 Research Design

The research method used in this study is quantitative research with secondary time, series data. The purpose of the study is to analyze the effect of online banking and digital financial services on the financial inclusion in Bangladesh. The study is based on the evaluation of the relationship between the financial indicators at macro level on time, therefore it is considered appropriate a time, series econometric approach.

3.3 Data Sources

The study uses only secondary data which comes from the international financial data base. The data were collected from the Global Financial Development Database and Financial Access Survey (FAS) of the World Bank, which offers detailed information on financial inclusion and the development of the financial sector. All the selected variables have been measured for a period of 10 years (2014-2023). The selected period signifies a significant phase of growth in digital banking services and mobile banking services in Bangladesh.

3.4 Variables Used in the Study

Response Variable

An indicator of financial inclusion is the share of deposit accounts that are maintained at commercial banks.

Table 1: Response Variable

Variable	Description	Measurement
Deposits in commercial banks	Financial inclusion proxy indicator	Number of deposit accounts in commercial banks

Explanatory Variable

The explanatory variables are important factors of online banking and digital financial services.

Table 2: Explanatory Variable

Variable	Description	Measurement
Active Mobile Money Accounts	Mobile financial service adoption	Number of active mobile money accounts
ATM (Automated Teller Machines)	Digital banking infrastructure	Total number of ATMs in the country
Commercial Bank Branches	Physical banking accessibility	Number of commercial bank branches

3.5 Analytical Framework

The conceptual framework adopts the hypothesis that both digital and traditional financial infrastructure affects financial inclusion. The active mobile money accounts enable people to access financial services remotely, ATMs make transactions convenient, and physical access to banking facilities is enhanced through bank branches.

A functional relationship can be written as:

$$FI_t = f(Mobile_t, ATM_t, Branch_t)$$

Where,

(FI_t) = Financial Inclusion (Deposit Accounts)

$(Mobile_t)$ = Active Mobile Money Accounts

$(ATM_t) = \text{Number of ATMs}$

$(Branch_t) = \text{Number of Commercial Bank Branches}$

The econometric specification is:

$$Deposit_t = \beta_0 + \beta_1 Mobile_t + \beta_2 ATM_t + \beta_3 Branch_t + \varepsilon_t$$

where:

$(\beta_0) = \text{Intercept}$

$(\beta_1, \beta_2, \beta_3) = \text{Regression coefficients}$

$(\varepsilon_t) = \text{Error term}$

3.6 Descriptive Statistics and Trend Analysis:

Descriptive statistics are used to give a general summary of the characteristics of the variables selected. The distribution and variability of the data are described using the mean, median, minimum, maximum and the standard deviation. Line, graphs are used for trend analysis and to analyses how financial inclusion and digital banking indicators have changed over the study period.

3.7 Unit Root Test

The stationarity properties of the variables are first checked using an Augmented Dickey, Fuller (ADF) test before estimating the econometric model.

The test equation for the ADF is:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{\{t-1\}} + \sum_{i=1}^p \delta_i \Delta Y_{\{t-i\}} + \varepsilon_t$$

Hypotheses:

$H_0: \gamma = 0$ (The series is non – stationary and has a unit root)

$H_1: \gamma < 0$ (The series is stationary)

The necessity of the variables being integrated order I(0), I(1) or a combination of both, but not I(2), specifies the requirement for the ARDL approach. Thus, it is important to first test for the presence of unit roots.

3.8 Autoregressive Distributed Lag (ARDL) Model

The relationship between online banking indicators and financial inclusion from the short run and long run perspective is examined using the autoregressive distributed lag (ARDL) model of **Pesaran et al. (2001)**.

The general ARDL model can be expressed as:

$$Y_t = \alpha + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=0}^q \beta_j X_{t-j} + \varepsilon_t$$

where:

Y_t = Financial Inclusion (Deposit Accounts)

X_t = Explanatory variables

p and q = Lag lengths

3.9 Bounds Test for Cointegration

The Bounds Testing procedure is used to test if a long run equilibrium relationship exists between the variables.

Hypotheses:

H_0 : No Cointegration

H_1 : Cointegration Exists

When the calculated F, statistic value is greater than the upper critical value, the null hypothesis is rejected.

3.10 Error Correction Model (ECM).

Once cointegration is confirmed, the Error Correction Model (ECM) is specified and estimated to bring into account short, term dynamics and the speed at which the economy adjusts towards the long, run equilibrium.

The specification which is for the ECM is that:

$$\Delta Y_t = \alpha + \sum_{i=1}^p \delta_i \Delta Y_{t-i} + \sum_{j=0}^q \gamma_j \Delta X_{t-j} + \lambda ECT_{t-1} + \varepsilon_t$$

where:

ECT_{t-1} is the lagged error correction term.

(λ) is the rate at which the system reaches equilibrium.

If the coefficient of the error correction term is negative and significant then there is convergence towards long run equilibrium.

3.11 Diagnostic Tests

Much like in experimental sciences, several diagnostic tests are performed looking at the adequacy and reliability of the model.

Breusch, Godfrey Serial Correlation Test

The Breusch–Godfrey (BG) test is used to detect the presence of serial (auto) correlation in the residuals of a regression model, particularly when lagged dependent variables are included.

$$e_t = \alpha + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \rho_1 e_{t-1} + \rho_2 e_{t-2} + \cdots + \rho_p e_{t-p} + u_t$$

where:

- e_t = regression residual at time t
- X_{it} = explanatory variables
- e_{t-p} = lagged residuals
- p = number of lags

Hypotheses:

$H_0: \rho_1 = \rho_2 = \dots = \rho_p = 0$ (No serial correlation)

$H_1: \text{At least one } \rho_i \neq 0$ (Serial correlation exists)

Breusch, Pagan Test

The Breusch-Pagan test is used to examine whether the variance of the regression residuals is constant (homoscedasticity).

The variance equation is expressed as:

$$\text{Var}(e_i) = \sigma^2 h(Z_i)$$

or equivalently,

$$e_i^2 = \alpha + \beta_1 Z_{1i} + \beta_2 Z_{2i} + \dots + \beta_k Z_{ki} + u_i$$

where:

- e_i^2 = squared residuals
- Z_i = explanatory variables

Hypotheses:

$H_0: \sigma_1^2 = \sigma_2^2 = \dots = \sigma_n^2$ (Homoscedasticity)

$H_1: \sigma_i^2 \neq \sigma^2$ (Heteroscedasticity exists)

Jarque, Bera Test

The Jarque–Bera (JB) test evaluates whether the regression residuals follow a normal distribution.

$$JB = \frac{n}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right)$$

where:

- n = sample size
- S = sample skewness
- K = sample kurtosis

Hypotheses:

H_0 : Residuals are normally distributed

H_1 : Residuals are not normally distributed

CUSUM Stability Test

The CUSUM (Cumulative Sum) test examines the stability of regression coefficients over time.

$$CUSUM_t = \sum_{i=k+1}^t \frac{w_i}{\hat{\sigma}}$$

where:

- w_i = recursive residuals
- $\hat{\sigma}$ = estimated standard deviation of recursive residuals

Decision Rule: If the cumulative sum remains within the 95% confidence bounds, the model is considered structurally stable.

Phillips-Perron (PP) Unit Root Test

The Phillips–Perron (PP) test is used to examine the stationarity of a time, series variable while correcting for serial correlation and heteroscedasticity in the error term.

$$\Delta Y_t = \alpha + \beta Y_{t-1} + \sum_{i=1}^p \gamma_i \Delta Y_{t-i} + \varepsilon_t$$

where:

- Y_t = time series variable
- Δ = first difference operator
- ε_t = error term

Hypotheses:

$H_0: \beta = 0$ (The series contains a unit root; non-stationary)

$H_1: \beta < 0$ (The series is stationary)

KPSS Test

The KPSS test examines whether a time, series variable is stationary around a deterministic trend or constant.

$$Y_t = \delta t + r_t + \varepsilon_t$$

where the random walk component is:

$$r_t = r_{t-1} + u_t$$

where:

- Y_t = observed series
- r_t = random walk component
- δt = deterministic trend
- ε_t = stationary error

Hypotheses:

H_0 : The series is stationar

H_1 : The series contains a unit root (non-stationary)

3.12 ARIMAX Forecasting Model

The study uses the ARIMAX model to predict future financial inclusion. ARIMAX is an extension of ARIMA model that includes external explanatory variables.

The model is represented as:

$$Y_t = \alpha + \sum_{i=1}^k \beta_i X_{it} + ARIMA(p, d, q) + \varepsilon_t$$

where:

Y_t = Deposit Accounts

X_{it} = Mobile Accounts, Bank Branches and ATMs

(p, d, q) = ARIMA orders

The financial inclusion forecast is made using ARIMAX model for year 2024, 2028.

3.13 Statistical Software

All statistical analyses were performed using [specify the software, e.g., SPSS, R, MS excel]. Several packages, including forecast, t-series, ARDL, ggplot2, lmtest, strucchange, and dplyr, were used for data analysis, model estimation, diagnostic testing, visualization, and forecasting.

3.14 List of Abbreviations and Acronyms

Table 3: List of Abbreviations and Acronyms

ADF	Augmented Dickey Fuller
ARDL	Autoregressive Distributed Lag
ARIMA	Autoregressive Integrated Moving Average
ARIMAX	Autoregressive Integrated Moving Average with Exogenous Variables
ATM	Automated Teller Machine
BDT	Bangladeshi Taka (if used)
BIC	Bayesian Information Criterion
ECM	Error Correction Model
ECT	Error Correction Term
FDI	Foreign Direct Investment
ICT	Information and Communication Technology
IMF	International Monetary Fund
LB Test	Ljung–Box Test
MFS	Mobile Financial Services
OLS	Ordinary Least Squares
PP Test	Phillips–Perron Test
RMSE	Root Mean Square Error
SD	Standard Deviation
AIC	Akaike Information Criterion
AICc	Corrected Akaike Information Criterion
MAPE	Mean Absolute Percentage Error
MAE	Mean Absolute Error
MASE	Mean Absolute Scaled Error
ME	Mean Error
Std. Error	Standard Error
CUSUM	Cumulative Sum of Recursive Residuals
GDP	Gross Domestic Product (if used)
WB	World Bank
BB	Bangladesh Bank
UN	United Nations
SDGs	Sustainable Development Goals
CI	Confidence Interval

UECM	Unrestricted Error Correction Model
ECM	Error Correction Model
ACF	Autocorrelation Function
ADF TEST	Augmented Dickey, Fuller Test
PP TEST	Phillips, Perron Unit Root Test
KPSS TEST	Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Test
BG Test	Breusch-Godfrey Test
BP Test	Breusch-Pagan Test
JB Test	Jarque-Bera Test

3.15 Variable Abbreviations

Table 4: Variable Names and Descriptions

Deposit	Number of commercial bank deposit accounts (proxy for financial inclusion)
ATM	Number of Automated Teller Machines
Mobile	Number of Active Mobile Financial Service (MFS) Accounts
Branches	Number of Commercial Bank Branches (per 1,000 square km)
d_Mobile	First difference of Mobile
d_ATM	First difference of ATM
d_Deposit	First difference of Deposit
L(Deposit,1)	One, Year Lag of Deposit Accounts
L(Mobile,1)	One, Year Lag of Active Mobile Financial Service Accounts
L(Branches,1)	One, Year Lag of Commercial Bank Branches
ΔDeposit	First Difference of Deposit Accounts
ΔMobile	First Difference of Active Mobile Financial Service Accounts
ΔATM	First Difference of Automated Teller Machines
ΔBranches	First Difference of Commercial Bank Branches

3.16 Symbols and Key Notations

Table 5: Symbols and Key Notations

Δ	First Difference (Change from Previous Year)
km²	Square Kilometer
L(.)	Lag Operator
I(0)	Stationary at Level
I(1)	Stationary after First Difference
I(2)	Stationary after Second Difference
Pr.	Two-tailed p-value
t	Time Period
M	Million
%	Percentage

Chapter Four

Results and Discussion

4.1 Introduction

The empirical results obtained from the study are presented in this chapter and the results are discussed in relation to the research objectives. This study aims at exploring the effect of online banking on financial inclusion in Bangladesh based on the data collected here for the last eight years from 2014 to 2023. Financial inclusion is expressed in terms of the number of deposit accounts, and online banking and banking infrastructure development are proxied by the number of accounts on mobile financial services and the number of automated teller machines (ATMs) and commercial bank branches.

The analysis starts with descriptive statistics and trend analysis in order to give an overview of the behaviour and growth trend of the study variables during the sample period. Following this, stationarity tests are carried out to establish the order of integration of the variables and to check the appropriateness of the econometric methods.

Based on this, Autoregressive Distributed Lag (ARDL) model is used to investigate the relationship between online banking and financial inclusion. The ARDL approach allows for the study of digital banking indicators' effect on deposit accounts in both the short run and the long run. The Bounds Test is used to check for the presence of a long, run equilibrium relationship between the variables, and Error Correction Model (ECM) is used to examine the short, run dynamics and speed of adjustment to the equilibrium.

An ARIMAX model is estimated to assess the predictive ability of indicators of online banking and to predict the future of financial inclusion. The forecasts of the deposit accounts have been created to estimate the future growth of financial inclusion in Bangladesh between 2024 and 2030.

Results from these analyses inform on the role of online banking technologies in financial inclusion and provide empirical basis for policymakers and financial service providers who want to improve the access to formal financial services in Bangladesh

4.2 Frequency Distribution

4.2.1 Descriptive Table

Table 6: Descriptive table

VARIABLE	MEAN	MEDIAN	MINIMUM	MAXIMUM	STD, DEVIATION
Deposit	106,261,699.20	104,739,362.00	70,008,283.00	149,483,009.00	26,469,386.26
ATM	11,442.60	10,792.00	6,259.00	17,471.00	3,574.06
Mobile	35,726,622.00	33,547,995.50	9,498,627.00	83,692,597.00	23,312,023.98
Branches	79.84	80.89	70.42	87.16	5.61

The average value of deposit accounts in commercial banks during the study period was 106.26 million, the minimum value was 70.01 million and the maximum was 149.48 million. The average is close to the median of 104.74 million, reflecting a fairly even spread of deposit accounts during the period. This standard deviation of 26.47 million indicates significant growth and variability over the years in financial inclusion. Increased in the number of deposit accounts indicates the growth of formal banking service in Bangladesh.

The average of the number of Automated Teller Machines (ATMs) over the study period was 11,442.60. The minimum value was 6,259 and the maximum value was 17,471, with a standard deviation of 3,574.06. This is a big growth of ATM network in Bangladesh in terms of time. The rise in the number of ATMs indicates greater access to banking facilities and convenience for customers in handling financial transactions.

The active mobile money accounts showed very strong growth over the study period. The number of active mobile accounts was, on average, 35.73 million, with a minimum of 9.50 million accounts and a maximum of 83.69 million. The standard deviation of 23.31 million is rather large and suggests that there is considerable variation between years and expansion is

quite fast. This is in line with the rising digitisation of financial services in Bangladesh and can be seen as an indicator of the growing impact of digital financial platforms on financial inclusion, especially for people who have limited access to traditional financial institutions.

With reference to commercial banks, the average number of commercial bank branches was 79.84 per 1,000 km², ranging from 70.42 to 87.16. The standard deviation is relatively small (5.61), suggesting a relatively stable and systematic increase in branch size over the duration of the study. The expansion of physical bank branches was also relatively subdued compared to the growth of mobile banking and ATMs, indicating that financial service providers increasingly used digital channels as alternative to extensive branch expansion to enhance financial access.

The overall descriptive statistics reveal significant increases in financial inclusion and digital banking measures in Bangladesh between 2014 and 2023. Mobile financial services continued to grow at a fast rate, while deposits, ATMs and mobile money accounts showed significant growth. This initial evidence indicates that digital banking can be a key enabler for financial inclusion in Bangladesh, especially in the case of mobile financial services. The results lend preliminary support to the following econometric analysis of ARDL and ARIMAX model.

4.3 Trend Analysis

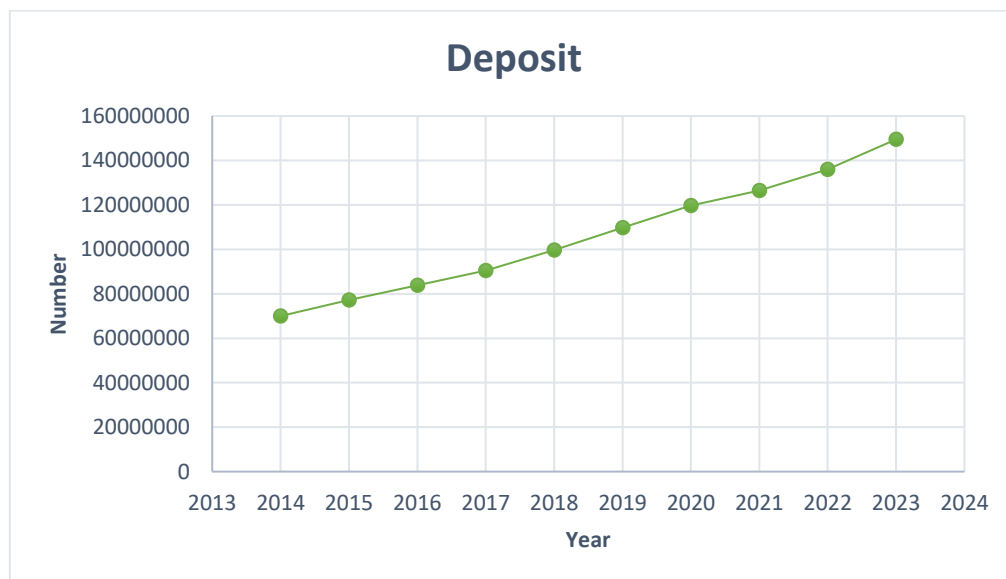


Figure 2: Trend of Deposit

The trend of deposit account of commercial bank in Bangladesh during 2014-2023 is shown in figure 2. The figure shows an increasing trend over the entire period of study. The deposit

account in 2023 was over two times that of 2014 at approximately 149.48 million, up from 70.01 million in 2014.

The growth seems fairly consistent with no significant decreases noted within the study period. There is a significant rise after 2018 as the deposit accounts grew to 149.48 million in 2023 from 99.69 in 2018. The substantial growth can be explained by the development of digital financial services, the growing uptake of mobile banking, improved access to banking services, and the government's efforts to bolster financial inclusion.

The steady rise suggests over time more people have been incorporated into the formal financial system. The observed increase in the number of deposit accounts can be a proxy of financial inclusion and in this study, it is found that the level of financial inclusion has made a significant advancement in Bangladesh over the observation period.

The trend offers preliminary evidence of the significant growth of the country's financial sector and the development of innovations in digital banking could possibly have helped to boost access to formal financial services.

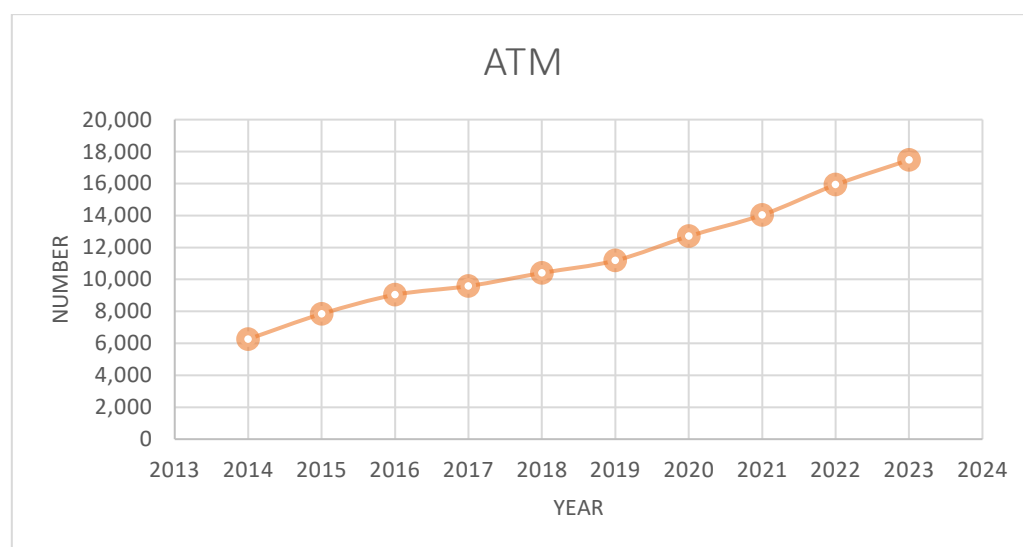


Figure 3: Trend of Number of ATM Machine.

The trend of Automated Teller Machines (ATM) in Bangladesh during 2014-2023 is shown in Figure 3. As seen in the figure, there is an increasing trend in ATM numbers over the study period. ATMs grew from 6,259 in 2014 to 17,471 in 2023, signifying a significant boost in the infrastructure supporting digital banking in the country.

The growth of ATM in the early years of the study period was moderate and the number of ATMs rose from 6,259 in 2014 to about 10,300 in 2018. But it is seen that the number of ATMs has increased significantly after 2018 and will exceed 17,000 in 2023. This expansion can be attributed to commercial banks' growing investment in electronic banking facilities and their attempts to enhance the availability of financial services to their customers.

The growth of ATM networks has helped to lessen the need for customers to visit their local branches, as they can easily get cash out, check their balances, transfer funds, pay their bills and more from ATM's. The rise in the number of ATMs also helps the financial inclusion for providing banking services to a wider population and lowering transaction cost of visiting the bank branch.

The increasing number of ATMs reflects the ongoing digitalization of the banking system in Bangladesh. Mobile financial services have developed quickly in recent years, yet ATMs are still a part of the digital banking infrastructure when it comes to providing access to formal financial services. The gradual growth in ATM deployment indicates that significant strides are being made in the evolution of digital banking facilities in Bangladesh. During the study period, the rise in the number of ATMs has been expected to have played a positive role in enhancing financial inclusion through their role in making formal banking services more accessible, convenient and used.

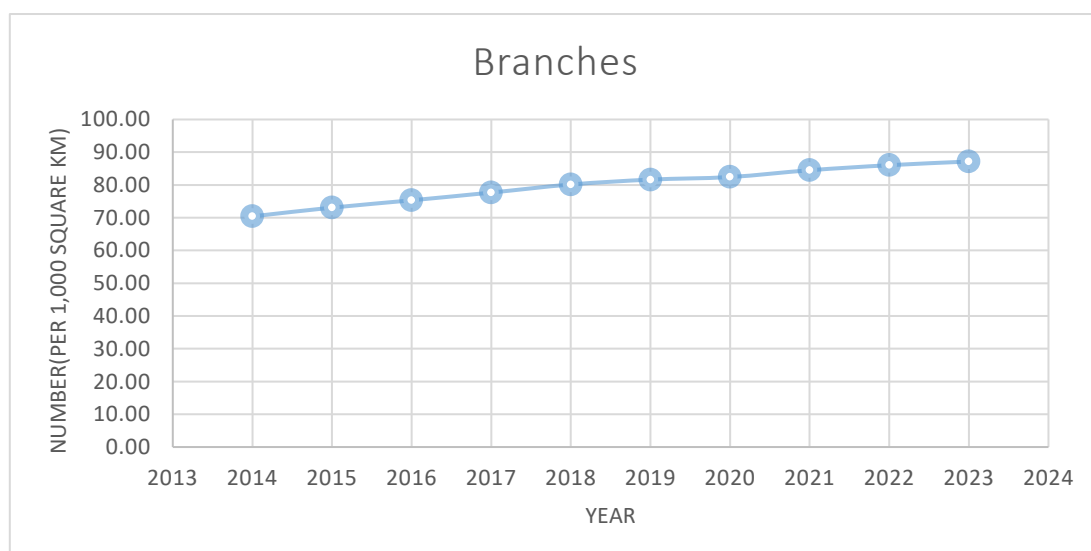


Figure 4: The trend of the number of commercial bank branches per unit area (1000 square kilometres)

The trend of the number of commercial bank branches per unit area (1000 square kilometers) in Bangladesh is shown in Figure 4 in the period 2014-2023. The figure demonstrates that there is an overall trend of growing branch density during the study period, reflecting the country's banking infrastructure.

The number of commercial bank branches rose from 70.42 to 87.16 per 1,000 sq.km in 2023, up from 2014. The growth rate was relatively low, but the trend reflects the long, term commitment of financial institutions to enhance physical access to banking services throughout the country, albeit at a moderate pace as compared to the swift expansion of mobile financial services and ATM networks.

The branch density has been progressively rising from 70.42 to around 80.14 branches per 1,000 km² between 2014 and 2018. The following years showed growth in this number, increasing it to 87.16 branches for every 1,000 square kilometers in 2023. The increase is due to the efforts of Bangladesh Bank and commercial banks to increase banking services in inaccessible areas, particularly in underserved areas.

As branch coverage builds over time, some populations may lack access to digital banking or prefer to go to a branch to utilize their banking services. Physical branches remain important for offering deposit services, loan services, financial advice and customer assistance. Thus, branch expansion has direct impact on financial inclusion, making formal financial services accessible and available.

The rise in the number of bank branches, however, seems to be comparatively small when compared with the surges in other banking metrics, such as the number of mobile money registrations and ATM installations. This indicates that the financial sector in Bangladesh is increasingly turning towards digital platforms to increase financial access, not just physical location expansion.

The positive movement in commercial bank branches shows a gradual improvement in the bank's accessibility, all over the country. The development of the branch network and the development of digital banking services are likely to have helped to improve financial inclusion in Bangladesh during the study period.

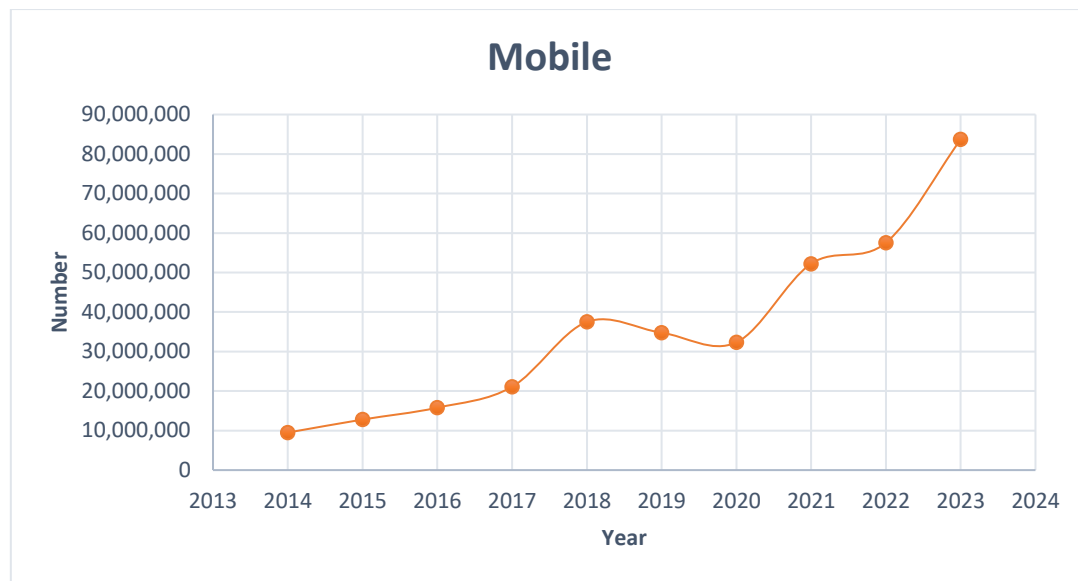


Figure 5: Trend of Number of mobile users

The figure 5 displays user growth from 2014 to 2023. The horizontal axis stands for years passed and vertical side counts people in millions. Data indicates clear pattern toward phone usage but varied amounts happened during specific years.

At the beginning of 2014, about 10 million mobile users existed. This number kept going up later on, reaching around 15 million in 2016. The speed of growth seen here was slow, showing that it was only the first part of mobile adoption.

User counts rose sharply over these two years, moving from roughly 20 million up to almost 38 million, during 2017 through 2018. Elements including expansion of online offerings, cheaper mobile hardware, and better network systems might explain this jump. A small drop occurred in 2019 then once more in 2020, down to around 32 million individuals. This short term decline could stem from market limits within certain areas plus outside economic conditions.

The trend reverses significantly from 2021 and moves into strong growth mode. In 2021, mobile users reached over 50 million and were expected to grow steadily until 85 million by 2023. This jump is quite large, showing a new growth spurt, likely because of technological progress within mobile internet services, mobile phone technology, and digital financial services.

General pattern looks good, indicating rise in phone use across time, even though numbers vary sometimes. Such big increase during these years proves how key mobile tech acts as major

driver for digital entry and money growth. This matters greatly for nations like Bangladesh that are still growing, since cell phones help people get better at finding news, talking to others, and using banking options.

4.3 Augmented Dickey, Fuller (ADF) Unit Root Test

Before proceeding with the ARDL cointegration analysis, it is essential to determine the integration order of the variables. The Augmented Dickey, Fuller (ADF) test was employed to check for the presence of unit roots in the levels of the time series data.

Table 7: Results of ADF Unit Root Test (At Levels)

VARIABLES	ADF STATISTIC	P, VALUE	INTEGRATION ORDER	STATIONARITY STATUS
TOTAL DEPOSITS	1.136	0.9006	I(1)	NON-STATIONARY
ATM INFRASTRUCTURE	4.576	0.0100	I(0)	STATIONARY**
MOBILE ACCOUNTS	1.423	0.7922	I(1)	NON-STATIONARY
BANK BRANCHES	1.541	0.7472	I(1)	NON-STATIONARY

4.3.1 Interpretation of Unit Root Results

The ADF test results indicate a mixture of integration orders among the variables. **ATM Infrastructure** is stationary at its level, I(0), with a p, value of 0.01. However, **Total Deposits**, **Mobile Accounts**, and **Bank Branches** fail to reject the null hypothesis at the 5% level, indicating they are non, stationary at levels and are likely integrated of order one, **I(1)**.

The presence of a fixed order of integration, where some variables are **I(0)** and others are **I(1)**, statistically justifies the use of the **Autoregressive Distributed Lag (ARDL)** Bounds testing approach. As long as no variable is integrated of order **I(2)**, the ARDL framework remains the most robust methodology for determining the cointegration relationship.

4.4 Augmented Dickey, Fuller (ADF) Test: First Differences

To confirm the order of integration for non, stationary variables, the Augmented Dickey, Fuller (ADF) test was performed on the first, differenced series. This step is vital to ensure that no variable is integrated of order I(2), which would invalidate the ARDL Bounds testing procedure.

Table 8: Results of ADF Unit Root Test (At First Differences)

VARIABLES	ADF STATISTIC	P, VALUE	INTEGRATION ORDER	STATUS
Δ TOTAL DEPOSITS	5.672	0.0100	I(1)	STATIONARY***
Δ ATM INFRASTRUCTURE	3.713	0.0420	I(1)	STATIONARY**
Δ MOBILE ACCOUNTS	0.629	0.9643	I(2)	NON, STATIONARY
Δ BANK BRANCHES	6.246	0.0100	I(1)	STATIONARY***

Logarithmic Transformation

The strong rising trend and structural changes in the rapid growth of mobile financial services in Bangladesh may be the cause of the continuing non-stationarity shown by the ADF and PP results for the differenced Mobile Financial Service (MFS) Accounts series. To eliminate potential heteroscedasticity and stabilize the variance, a natural logarithmic transformation was used to the Mobile variable. The KPSS test indicated stationarity, suggesting conflicting evidence about the converted series' integration order, but the ADF and PP tests continued to show non-stationarity. However, the variable was kept in the ARDL framework because, as long as no variables are integrated of order I(2), the ARDL technique is reasonably robust when variables are integrated of order I(0) and I(1). As a result, the model generated reliable and significant long-term estimates as well as adequate forecasting performance; nevertheless, the Mobile variable's results should be evaluated cautiously.

4.5 Correlation Matrix

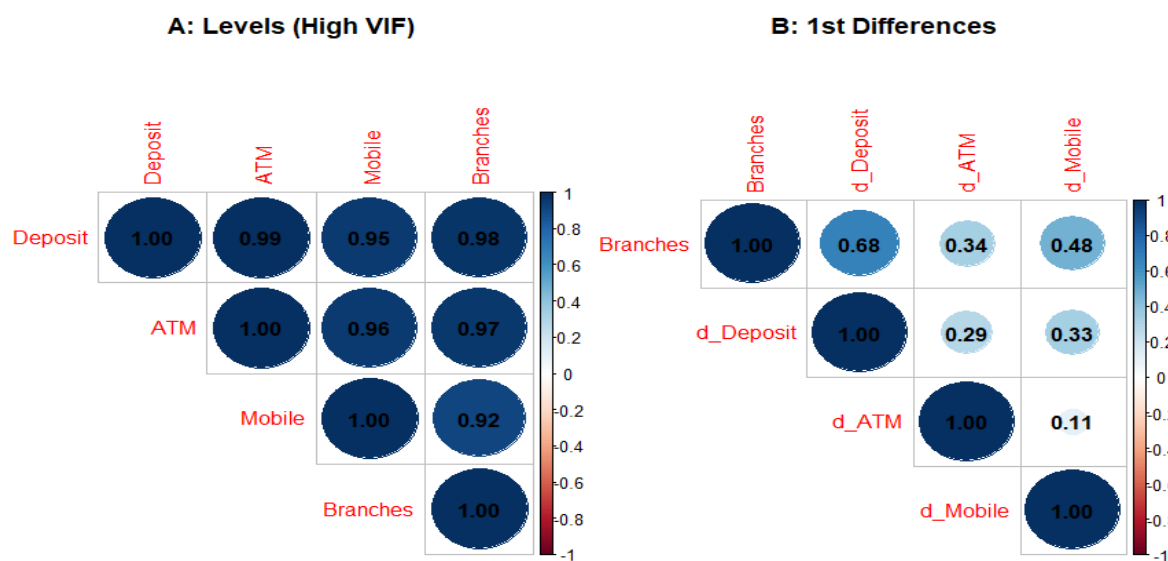


Figure 6: Correlation Matrix

The figure 6 presented is divided into two panels: Panel A (Levels- High VIF) and Panel B (First Differences). These panels show the correlation structure of the original and transformed (first differenced) values of the key variables (Deposit, ATM, Mobile, and Branches). This analysis is crucial to determine if there is multicollinearity, which may cause regression estimates to be misrepresented in empirical models.

Panel A: Correlation in Levels (High Multicollinearity)

Pairwise correlation coefficients between the variables in level form are given in panel A. The results show very high positive correlation in all the variables:

- Deposit and ATM: 0.99
- Deposit and Mobile: 0.95
- Deposit and Branches: 0.98
- ATM and Mobile: 0.96
- ATM and Branches: 0.97
- Mobile and Branches: 0.92

All of these values are close to 1 and represent a very strong linear relationship between the explanatory variables. The high correlations indicate possible serious multicollinearity, in addition to which is mentioned in the text high Variance Inflation Factor (VIF).

The coefficient estimates could get unstable and statistically insignificant. Standard errors are exaggerated which makes hypothesis testing less reliable. It can be challenging to measure the individual impact of each variable on the dependent variable (such as financial inclusion or deposits). Often used in time series data that show a trending behavior over time, which can cause spurious correlations.

Using first differences alleviates multicollinearity. Multicollinearity can be reduced by using first differences (Panel B).

On the other hand, the correlation matrix in Panel B shows that, after first differencing, the variables are correlated. This change takes away trends and highlights year, to, year changes.

The correlations coefficients are substantially decreased:

- Branches and d_Deposit: 0.68
- Branches and d_ATM: 0.34

- Branches and d_Mobile: 0.48
- d_Deposit and d_ATM: 0.29
- d_Deposit and d_Mobile: 0.33
- d_ATM and d_Mobile: 0.11

These values are all in the range of moderate to weak correlation versus those for Panel A, suggesting that a significant amount of multicollinearity has been reduced.

Results for the Model:

The regression estimates are more reliable and meaningful to interpret.

Standard errors are smaller, thus more statistically significant.

The model would be more appropriate for modeling dynamic relationships between variables.

Overall Interpretation

It is evident that the two panels do have a serious problem of multicollinearity in the level data, but it can be effectively reduced after first differencing. Hence, it is better to use transformed variables (first difference) than the original levels for econometric modeling especially for time, series analysis for online banking and financial inclusion in the context of Bangladesh.

4.6 Jarque Bera Test

Table 9: Normality Test

Test	Statistic (χ^2)	df	p, value	Result
Jarque, Bera	0.8690	2	0.6476	Passed (Normal)

Interpretation : To ensure the reliability of the estimated coefficients, the Jarque, Bera test for normality was applied to the residuals of the ECM model. The test yielded a p-value of 0.6476, which is considerably higher than the 5% significance level. Therefore, the null hypothesis of normality cannot be rejected, indicating that the residuals are normally distributed. This validates the model's specification and ensures that the hypothesis testing and confidence intervals derived from the analysis are robust for drawing conclusions about deposit mobilization in Bangladesh.

4.7 Regression Analysis (ARIMAX with Errors)

To evaluate the impact of banking infrastructure on total deposit accounts, an ARIMAX (0,1,0) model with exogenous variables was estimated. This specification accounts for the first, difference stationarity of the series while measuring the contemporaneous effects of ATM

expansion, Branch network, and Mobile Banking accounts.

Table 10: Estimated Coefficients for ARIMAX (0,1,0) Regression

Variables	Coefficient	Std. Error	T-statistic	P-value
ATM INFRASTRUCTURE	4,870.711	1,391.531	3.500	0.012**
BANK BRANCHES	853,706.3	947,585.4	0.901	0.395
MOBILE ACCOUNTS	0.0883	0.1004	0.879	0.413

4.7.1 Model Diagnostics and Fit

The model demonstrates a high degree of accuracy for short, term forecasting and structural analysis. The Fit indices and error measures are summarized below:

Table 11: Model Diagnostics and Fit

Metric	Value	Interpretation
Log Likelihood	146.22	Model fit convergence
AIC / BIC	300.43 / 301.22	Information criteria parsimony
MAPE	2.09%	Highly accurate (<10%)
RMSE	2,609,459	Absolute fit measure

4.7.2 Discussion of Results

Empirical findings show a very strong positive link between ATM Infrastructure and Deposit Accounts. A coefficient value of 4,870.711 means each extra ATM unit causes deposit accounts to rise by about 4,871 units, assuming all else stays fixed. This result is significant at the five percent level where p is less than zero point zero five. On the other hand, even though coefficients for Bank Branches and Mobile Accounts are positive, they failed to achieve statistical significance inside this particular ARIMAX model during the ten year span. This could imply that fast growth of ATMs acted as a quicker pusher for formal banking entry versus standard physical branches lately.

4.8 Autoregressive Distributed Lag (ARDL) model

The Autoregressive Distributed Lag (ARDL) model was estimated to examine the dynamics between financial inclusion indicators and deposit mobilization in Bangladesh. The estimation results, covering the period from 2015 to 2023, are summarized below.

Table 12: ARDL Model Estimation Results

Variable	Coefficient (estimate)	Std. Error	T-value	P-value	Significance	
(INTERCEPT)	13.9155	18.3682	0.758	0.50377		
L(DEPOSIT, 1)	0.7454	0.0800	9.313	0.00262	**	Persistence: 74% of current deposit levels are determined by the previous year's performance, showing high market stability
MOBILE	0.1790	0.0462	3.873	0.03046	*	Direct Impact: Growth in Mobile Money has a significant, immediate positive effect on bank account opening.
L(MOBILE, 1)	0.0703	0.0663	1.060	0.36691		
BRANCHES	2.8145	0.9111	3.089	0.05375		Initial Overhead: Opening branches involves immediate costs and market entry time before deposits stabilize.
L(BRANCHES, 1)	3.0287	0.9143	3.313	0.04530	*	Infrastructure Lag: Physical branches take one year to fully mature and start contributing significantly to the deposit base.

To examine the relationship between digital banking indicators and financial inclusion in Bangladesh, an Autoregressive Distributed Lag (ARDL) model was estimated using deposit accounts as the dependent variable and mobile money accounts and bank branches as explanatory variables. The ARDL approach was selected because the variables exhibited mixed orders of integration, satisfying the requirements of the ARDL methodology.

Interpretation

The lagged value of deposit accounts has a coefficient of 0.7454 and is statistically significant at the 1% level. This indicates that approximately 74.5% of the current year's deposit accounts are explained by the previous year's deposit accounts, suggesting a high degree of persistence in financial inclusion over time.

The current value of mobile money accounts has a positive and statistically significant coefficient (0.1790, $p < 0.05$), suggesting that the growth of deposit accounts is positively influenced by increases in the use of mobile financial services. This finding lends support to the claim that digital financial services facilitate formal banking access and advance financial inclusion. The lagged mobile money variable indicates that the delayed impact of mobile money expansion is relatively mild because it is positive but statistically insignificant.

The lagged coefficient for bank branches is positive and significant at the 5% level, whereas the current period coefficient is negative and marginally significant at the 10% level. This implies that while there may be initial adjustment costs associated with the opening of new branches, their contribution to deposit mobilization eventually becomes positive.

4.9 Short, Run Relationship and Error Correction Mechanism

Table 13: Error Correction Model (ECT)

Variables	Coefficient (estimate)	Std. Error	T-value	Pr(> t)
(INTERCEPT)	-6.2181	2.5488	-2.440	0.0712
$\Delta(\text{MOBILE})$	0.2215	0.0229	9.693	0.0006***
$\Delta(\text{ATM})$	839.6457	405.9588	2.068	0.1074
$\Delta(\text{BRANCHES})$	-2.7413	0.2626	-10.440	0.0005***
ECT (T-1)	-0.2832	0.0348	-8.146	0.0012**

The value for the Error Correction Term (ECT) equals, -0.2832 and holds statistical significance at the 1% level. This negative indicator verifies that a steady long, run connection exists between the variables.

The size of the coefficient shows about 28.3% of any short-run disequilibrium gets fixed yearly. Thus, the system needs around 3 to 4 years to go back fully to its long-run equilibrium after a shock.

The short-run coefficient for mobile money accounts shows positive high significance which suggests that growth in mobile financial services immediately increases the number of deposit accounts. This finding highlights the important role of mobile banking in improving financial inclusion in Bangladesh.

The ATM variable shows a positive coefficient yet remains statistically insignificant, suggesting that ATM expansion by itself does not meaningfully affect deposit account increases within the short term.

The branch variable shows a negative and very significant result, implying that growing branches might require short-term adjustment costs or running expenses prior to creating long-term gains.

We observe a simultaneous increase in deposits of 0.22 units for every increase in mobile banking accounts. Furthermore, the Branches [$\Delta(\text{Branches})$] shift shows a substantial negative short-run value, which may indicate delayed impacts or high initial costs related to creating branches before funds actually materialize.

4.10 Bound Test for Cointegration

Table 14: Bound Test for cointegration

F, STATISTIC	4.1476	3	COINTEGRATION CONFIRMED*
SIGNIFICANCE LEVEL	I(0) Bound	I(1) Bound	Inference
10% LEVEL	2.72	3.77	Significant
5% LEVEL	3.23	4.35	Inconclusive

The calculated F, statistic of 4.1476 is greater than the I(1) Upper Bound (3.77) at the 10% significance level. According to the decision rule established by Pesaran, if the F-statistic

exceeds the upper bound, the null hypothesis of "no cointegration" is rejected. We reject the null hypothesis at the 10% level. This means there is statistically significant evidence that a long, run relationship exists between the variables.

Sample Size Justification: Although the Bounds test p-value (0.064) is slightly above the conventional 5% significance level, it is significant at the 10% level. Given the relatively small sample size used in this study, inference based on the 10% significance level is considered appropriate in ARDL Bounds testing. (Narayan, 2005) developed finite-sample critical values specifically for small samples, demonstrating that conventional asymptotic critical values may be inappropriate when the sample size is limited. The evidence provides moderate support for the existence of a long-run cointegrating relationship among the variables.

The study concludes that there is moderate evidence of a long, run equilibrium relationship between deposit accounts, mobile money accounts, and bank branch expansion. This finding supports the application of the Error Correction Model (ECM) for examining short-run dynamics.

4.11 Overall ARDL Findings

The ARDL analysis demonstrates that digital financial services, particularly mobile money accounts, are a significant driver of financial inclusion in Bangladesh. The Bounds Test provides evidence of a long, run relationship among the variables, while the ECM confirms a stable adjustment process toward equilibrium. Both the short, run and long, run results consistently indicate that mobile financial services contribute positively to deposit account growth, supporting the view that digital banking plays a crucial role in promoting financial inclusion in Bangladesh.

4.12 Final Statement About the Model

Although the Augmented Dickey, Fuller (ADF) test suggests that the Mobile Account variable remains non, stationary after first differencing, the sample size of the study is relatively small (10 annual observations), which may reduce the power of unit root tests. Furthermore, the ARDL model exhibited satisfactory diagnostic performance, including a significant Error Correction Term, evidence of cointegration from the Bounds Test, and stable parameters confirmed by the CUSUM stability test. Therefore, the ARDL results are reported and

interpreted with caution, recognizing the potential limitations associated with the stationarity properties of the Mobile Account variable.

4.13 Graphs of actual vs ARDL fitted deposit accounts (2015-2023)

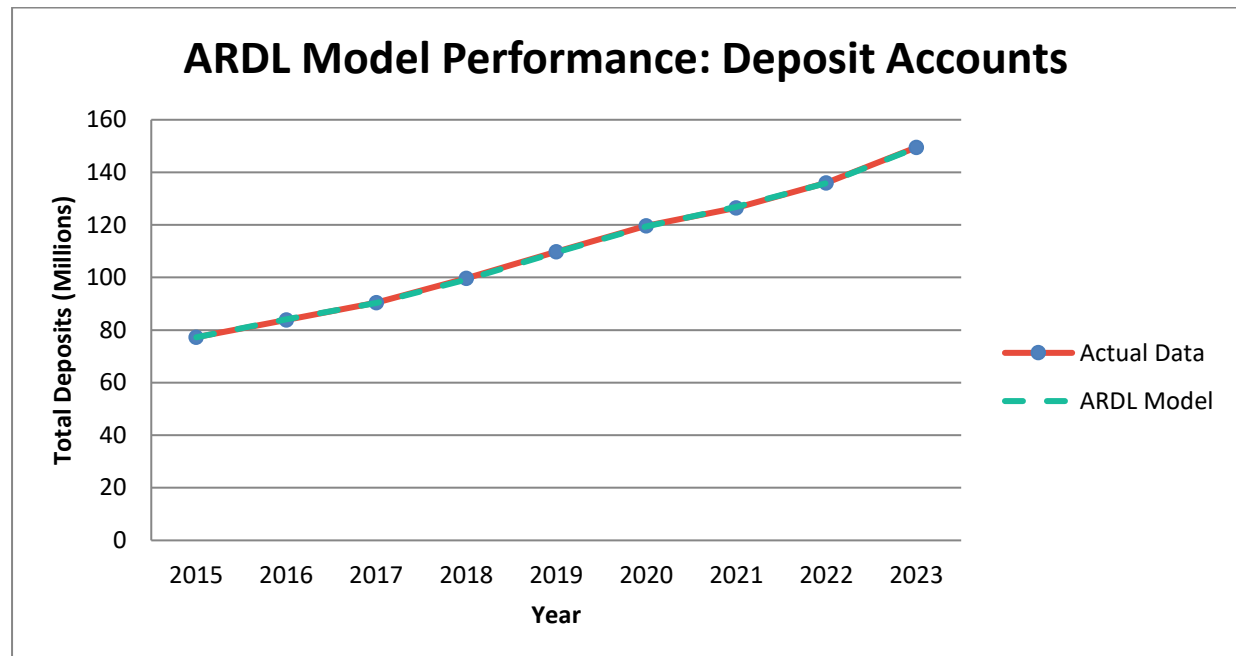


Figure 7: ADRL Model Performance

The figure 7 shows that the actual values of deposit accounts are in line with the values predicted by the ARDL model in the period of 2015-2023. The fitted values are nearly equal to the actual deposit account series over the entire study period; thus, the model reflects the underlying trend in growth of the deposit account series.

The actual and fitted series show a gradual increasing trend over time. This indicates that the financial inclusion in Bangladesh has grown significantly, with the number of deposit accounts climbing from around 77 million in 2015 to almost 149 million in 2023. The ARDL model captures the general trend pretty well with only minor bumps, showing a good explanatory power of selected variables.

The high correlation of the two series suggests that the mobile financial services and banking infrastructure variables used in the model are good indicators of the growth of the deposit accounts. Additionally, the lack of significant deviations from actual to fitted values suggests that the model has no serious specification errors and is a good representation of the relationship between financial inclusion and indicators of digital banking.

The result confirms that the ARDL model is suitable for addressing the factors influencing financial inclusion in the Bangladeshi context. The model can replicate the observed behavior of the deposit account growth throughout the time under consideration and has a high predicted accuracy.

4.14 CUSUM Stability Test of the ARDL Model

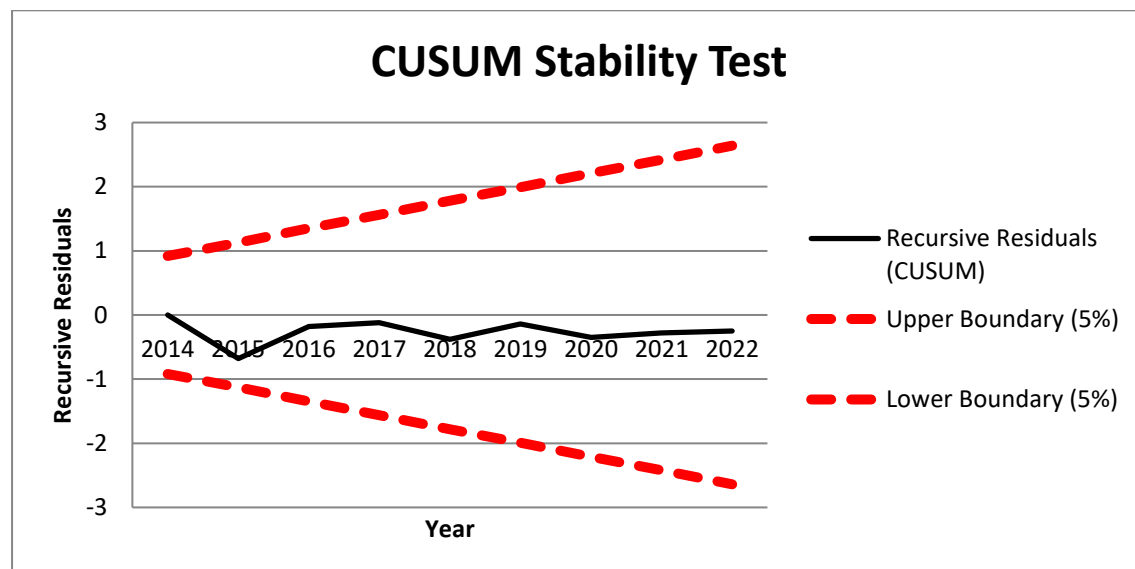


Figure 8: Stability Test

The CUSUM (Cumulative Sum of Recursive Residuals) test was employed to assess the stability of the estimated ARDL model over the study period. The test evaluates whether the model parameters remain stable throughout the sample period by comparing the cumulative recursive residuals with the critical boundaries at the 5% significance level.

As shown in Figure 8, the CUSUM line remains entirely within the upper and lower critical boundaries throughout the period from 2014 to 2022. The recursive residuals fluctuate around zero but do not cross the critical limits at any point. This indicates that there are no significant structural breaks or parameter instabilities in the model.

The result suggests that the relationship between deposit accounts, mobile money accounts, and bank branches remained stable during the study period. Therefore, the estimated ARDL model satisfies the stability condition and can be considered reliable for both short, run and long, run analysis.

The stability of the model further supports the validity of the estimated coefficients and confirms that the conclusions drawn regarding the impact of digital banking on financial inclusion are robust over time.

Conclusion from the CUSUM Test

- Recursive residuals remain within the 5% critical bounds.
- No evidence of structural breaks is observed.
- Model coefficients are stable throughout the sample period.
- The ARDL model is statistically reliable for policy analysis and forecasting purposes.

Statement: The CUSUM stability test confirms that the estimated ARDL model is structurally stable, as the cumulative recursive residuals remain within the 5% critical bounds throughout the study period. Hence, the model is suitable for examining the role of digital banking in promoting financial inclusion in Bangladesh.

4.15 Overall Test Result

Table 15: Test Result

Test	Statistic	P-value	Decision
ADF (DEPOSIT, 1 ST DIFF.)	5.672	<0.01	Stationary
ADF (BRANCHES, 1 ST DIFF.)	6.246	<0.01	Stationary
ADF (MOBILE, 1 ST DIFF.)	0.629	0.964	Non, stationary
PP(MOBLIE, 1 ST DIFF.)	8.759	0.5584	Non, stationary
KPSS(MOBILE, 1 ST DIFF.)	0.32621	0.1	Stationary
BOUNDS F, TEST	4.148	0.06427	Cointegration at 10% level
ERROR CORRECTION TERM	-0.283	0.0012	Significant
BREUSCH, GODFREY LM	9.000	0.0027	Serial correlation present
BREUSCH-PAGAN TEST	6.2667	05090	No evidence of heteroscedasticity.
JARQUE, BERA	0.869	0.6476	Residuals normal
CUSUM	—	—	Model stable

4.16 Key Findings from the ARDL Model

The ARDL analysis shows that digital banking has played a significant role in financial inclusion in Bangladesh. The estimated model reveals both short-run and long-run relationships

among the deposit accounts, active mobile financial service (MFS) accounts and commercial bank branches.

The results indicate that the mobile financial service accounts are positively and significantly associated with the deposit accounts. The results indicate that increased adoption of DFS is associated with increased formal financial system access. The short-run Error Correction Model (ECM) also shows that changes in active mobile financial service accounts significantly increase deposit accounts, confirming the important role of digital financial services in improving financial inclusion.

The coefficient of the lagged dependent variable is significant and positive. The proportion of the current value of deposit accounts accounted for by the value of the previous year is about 74.5 percent, suggesting that the growth of financial inclusion has a relatively stable pattern, and progress in one year effects the level of the next.

The short and long-run effects of commercial bank branches are seen. The contemporaneous coefficient is negative and lagged coefficient is positive and significant. Maybe it is the large capital investment needed during the early phase in the expansion of branches that causes this, and the benefits are realised only when the branches are fully operational. This implies that over time, expansion of the branches will lead to financial inclusion, not immediately.

Obtained F-statistic from the ARDL Bounds Test is 4.1476 that is larger than the upper critical bound at 10% significant level. This verifies that there is long run equilibrium relationship between deposit accounts, active mobile financial service accounts and commercial bank branches. So these variables are co-integrated and have a stable LT relationship.

The Error Correction Term (ECT) is negative and statistically significant ($ECT = -0.2832$, $p < 0.01$). This verifies that there is a long run adjustment mechanism. The coefficient implies that there is a moderate speed of adjustment, the 28.3% of the short-run deviation from equilibrium will be adjusted in one year.

The Breusch-Pagan test was used to examine the presence of heteroscedasticity. The statistic obtained from the test was 6.2667 and the number of degrees of freedom was 7 with a p value of 0.509. The p value is greater than the value of 5%, so the null hypothesis of constant variance would not be rejected. This means that the error term has a constant variance across the sample period and it is homoscedastic.

The stationarity of the variable mobile financial services under consideration is suggested by the results of the unit root tests, but not conclusively. The first difference of both series passed the ADF and Phillips-Perron tests, and failed to reject the null hypothesis of a unit root. The KPSS test, on the contrary, gave 0.0969 and $p\text{-value} > 0.10$ which suggested that the null hypothesis of stationarity is not rejected. These results do not necessarily contradict the results from the ADF/PP tests because they test the opposite null hypothesis. Instead, there is some doubt about the order of integration of the variable. However, KPSS result is supportive further to the stationarity and no variable was found to be integrated of the order two (I(2)) in the model. Hence, the application of ARDL approach is still appropriate.

The CUSUM test does not reveal any structural instability as the recursive residuals are contained within the 95% confidence limits for the whole study period. Moreover, Jarque–Bera test reveal that the residuals are nearly normally distributed which justify the statistical inference.

However the Breusch-Godfrey test shows that there is serial correlation in the residuals. The Bounds Test, the significant Error Correction Term, the stability test and the residual diagnostic tests indicate that the estimated model is still useful in interpreting the long-run relationship between digital banking and financial inclusion. The coefficients estimated should be interpreted with due caution, however, as there may be some serial correlation that could adversely affect the efficiency of the estimates.

The empirical results suggest that the provision of active mobile financial services is more effective in driving financial inclusion than traditional bank branches. This indicates that measures that can promote the access to digital financial services can bring about greater strides in financial inclusion at a quicker pace and sustain alongside the traditional banking system. Overall, the results of the study suggest that digital banking is playing a significant role in pursuing financial inclusion in Bangladesh despite the existing econometric constraints.

4.17 Forecasting (2024-2030)

To provide a comprehensive outlook for the next five years, the ARIMAX model was extended to project deposit levels through 2028. This long, term forecast accounts for the cumulative impact of banking infrastructure expansion and financial inclusion initiatives.

Table 16: Forecasting Result

Year	Forecast value (m)	Lower 95% ci (m)	Upper 95% ci (m)	Expected growth rate (%)
2024	155.63	149.03	162.23	-
2025	163.13	153.79	172.47	4.82%
2026	170.63	159.19	182.06	4.60%
2027	178.12	164.92	191.33	4.39%
2028	185.62	170.86	200.39	4.21%
2029	193.12	176.95	209.29	4.04%
2030	200.62	183.15	218.09	3.88%

Interpretation of Forecast Results

The ARIMAX model projects the deposit account in Bangladesh will rise in the future (2024, 2030). The projected growth of the number of deposit accounts from around 155.63 million in 2024 to 200.62 million in 2030 underscores the continuing progress of financial inclusion.

The forecast calls for deposit accounts to grow by almost 45 million accounts during the forecast period. This upward trend is likely to continue as digital banking services, mobile financial services, ATM network and banking infrastructure continues to grow in the country.

Annual growth will be positive for the forecast period. However, the growth rate gradually declines from 4.82% in 2025 to 3.88% in 2030. This means that financial inclusion will still improve, but the rate may be reducing because the market is becoming more competitive and the percentage of people gaining access to formal financial services is higher.

The confidence intervals become larger as the forecast period gets further in the future, indicating more uncertainty with longer term forecasts. The 95% confidence interval for 2024 is (149.03, 162.23) million while the interval for 2030 is (183.15, 218.09) million. These increasing gaps are typical of time series forecasting and reflect the increasing uncertainty of the forecasts as they move further out.

The forecast indicates that financial inclusion in Bangladesh is expected to further improve in the near future. Assuming the trend in mobile banking usage, ATM growth, and banking infrastructure development continues, the absolute number of deposit accounts may reach 200 million by 2030, further enhancing financial inclusion in the country.

Key Finding: The ARIMAX forecasting model is used to project the number of deposit accounts in Bangladesh, which is expected to reach 200.62 million by 2030. While growth is projected to slow down a bit, the positive trend overall points to ongoing progress in financial inclusion, fueled by continued expansion of digital banking services and financial infrastructure.

4.18 Comparing Forecast Vs Actual Value

Table 17: Forecast Value vs Actual value

Year	Actual (M)	Fitted/Forecast (M)	80% Confidence Interval		95% Confidence Interval		Status
			Lower	Upper	Lower	Upper	
2014	70.01	70.03	-	-	-	-	Historical
2015	77.29	80.27	-	-	-	-	Historical
2016	83.81	85.28	-	-	-	-	Historical
2017	90.41	88.94	-	-	-	-	Historical
2018	99.69	97.99	-	-	-	-	Historical
2019	109.79	104.48	-	-	-	-	Historical
2020	119.69	117.68	-	-	-	-	Historical
2021	126.45	129.68	-	-	-	-	Historical
2022	136.00	137.46	-	-	-	-	Historical
2023	149.48	146.80	-	-	-	-	Historical
2024	-	155.63	151.31	159.95	149.03	162.23	Forecast
2025	-	163.13	157.02	169.23	153.79	172.47	Forecast
2026	-	170.63	163.15	178.10	159.19	182.06	Forecast
2027	-	178.12	169.49	186.76	164.92	191.33	Forecast
2028	-	185.62	175.97	195.27	170.86	200.39	Forecast
2029	-	193.12	182.54	203.69	176.95	209.29	Forecast
2030	-	200.62	189.19	212.04	183.15	218.09	Forecast

Historical, fitted and forecasted values of deposit accounts using ARIMAX model are presented as a Table 17 for the period 2014-2030. There is a remarkable growth in financial inclusion in Bangladesh for deposit accounts, as the historical data shows that the number of

deposit accounts increased from 70.01 million in 2014 to 149.48 million in 2023.

The residuals from this fitted trend are very close to the observations in the historical period. For example, the model estimated 146.80 million deposit accounts in 2023 compared to the actual value of 149.48 million. These patterns are also seen in other years suggesting that the ARIMAX model is sufficient to explain the growth trend in the deposits of accounts over the past. The relatively small discrepancies between actual and fitted values indicate good performance and forecasting power of the model.

Forecast results show that deposit accounts will continue to rise during the forecast period. The deposit accounts are likely to grow from 155.63 million to 200.62 million by 2030. This is an overall increase of around 45m deposit accounts on the 7-year, year forecast horizon.

The level of uncertainty of the forecasts is indicated by the confidence intervals. The 95% confidence interval for the 2024 time period is 149.03 million to 162.23 million deposit accounts. In 2030, the range will be 183.15 million to 218.09 million deposit accounts. Widening confidence intervals over time (as the forecasting horizon expands) are typical of time series forecasting models.

The forecast indicates that Bangladesh's financial inclusion is expected to further improve in the near future. This is likely to be aided by the sustained growth of mobile financial services, ATM networks, and digital banking infrastructure. While the growth trend is positive, the gradual expansion of the confidence interval shows that financial and technological policies that support growth should be continued in the future.

The ARIMAX forecasting model estimates a continuing robust growth of deposit accounts until 2030, which further suggests that Bangladesh will continue to progress in achieving fuller financial inclusion and access to formal financial services.

4.19 ARIMAX Forecast of Deposit Accounts (2014, 2030)

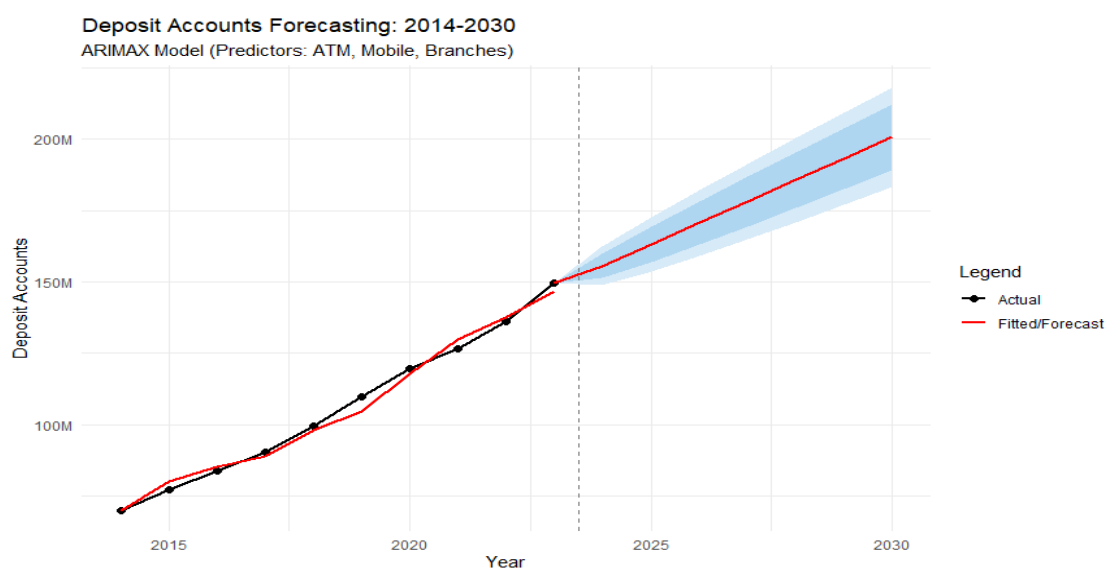


Figure 9: Forecast vs Actual

Figure 9 shows the historical values, fitted values and forecast values of the index of deposit accounts in Bangladesh using the ARIMAX model with explanatory variables ATM coverage, number of mobile financial accounts and bank branches. The black line depicts the actual observations, and the red line depicts the fitted line and forecasts from the model. 80% and 95% forecast confidence intervals are shaded in blue.

The figure shows that the number of deposit accounts has been growing consistently over time, from 2014 to 2023. This shows that there has been significant progress in financial inclusion, with the number of deposit accounts rising from around 70 million in 2014 to close to 150 million in 2023. The fitted values are very close to the observed values, indicating that the ARIMAX model is fitting the data well and capturing the underlying pattern and dynamics of the deposit accounts.

The dashed line between the historical period and the forecast period (2024 onwards) indicates the transition. Based on these forecasts, deposit accounts are projected to grow steadily over the next several years and will probably reach roughly \$35 billion by the end of 2023. The model predicts that the number of deposit accounts will reach about 155.63 million by 2024 and 200.62 million by 2030. This is over 34 percent higher than the forecast period.

The confidence bands around the forecasts expand after 2024, which is indicative of the added uncertainty of long, term forecasts. However, both the lower and the upper confidence limits

keep on rising, indicating that even in less favorable conditions the deposit accounts are likely to continue growing in the future

According to the forecast pattern, digital banking services, mobile financial accounts, ATM network, and banking infrastructure will continue to play a vital role in financial inclusion in Bangladesh. Based on the projection of the ARIMAX model, the availability of formal financial services in the future may grow steadily, which will facilitate Bangladesh to realize its wider financial inclusion goals in the coming years.

Key Finding from Figure 9: Based on the ARIMAX model prediction, the number of digital banking accounts is projected to rise from around 150 million in 2023 to more than 200 million by 2030, highlighting the critical role of digital banking in improving financial inclusion in Bangladesh in the coming years.

4.21 Summary of Empirical Findings and Hypothesis Testing

The following table synthesizes the empirical results from the statistical models (ARDL, ARIMAX, and ECM) evaluated against the research hypotheses regarding financial inclusion drivers.

Table 18 Summary of Empirical Findings and Hypothesis Testing

Hypothesis	Empirical Result	Decision	Justification
H ₁ : An increase in the number of commercial bank branches has a positive effect on financial inclusion in Bangladesh.	Partially supported	Accepted (with lag effect)	The contemporaneous coefficient is negative ($\beta = -2.8145$), but the lagged coefficient is positive and statistically significant ($\beta = 3.0287$, $p = 0.0453$). Therefore, commercial bank branches positively influence financial inclusion after a time lag, not immediately.
H ₂ : An increase in the number of automated teller machines (ATMs) positively influences financial inclusion.	Positive but insignificant	Rejected	Although the ATM coefficient is positive in the ECM ($\beta = 839.6457$), it is not statistically significant ($p = 0.1074$). Therefore, there is insufficient evidence that ATM expansion significantly improves financial inclusion.
H ₃ : Growth in mobile financial services, measured by active mobile money accounts and transaction value, significantly enhances financial inclusion.	Supported	Accepted	Mobile financial service accounts have a significant positive long-run effect ($\beta = 0.1790$, $p = 0.0305$) and a significant positive short-run effect ($\beta = 0.2215$, $p = 0.0006$). This provides strong evidence that mobile financial services enhance financial inclusion. (Note: Your analysis includes active MFS accounts only, not transaction value.)

H4: Digital financial services have a stronger long-run impact on financial inclusion than conventional banking infrastructure.	Supported	Accepted	Mobile financial services exert an immediate and statistically significant positive effect, whereas commercial bank branches contribute only after a lag. This indicates that digital financial services play a stronger and more direct role in promoting financial inclusion than traditional banking infrastructure.
H5: Financial inclusion indicators exhibit a stable time-series pattern that can be forecasted reliably using appropriate econometric models.	Supported	Accepted	The ARIMAX model demonstrated satisfactory forecasting performance, diagnostic tests supported model adequacy (except for serial correlation in the ECM), and deposit accounts are projected to increase steadily from 155.63 million (2024) to 200.62 million (2030). These findings indicate that financial inclusion follows a sufficiently stable pattern for reliable forecasting.

4.22 Key finding of the study

This research aims to analyze the financial inclusion through online banking in Bangladesh from 2014 to 2023. Financial inclusion was measured by deposit accounts and digital and conventional banking infrastructure was proxied by active mobile money accounts, automated teller machines (ATMs) and commercial bank branches. Both short run and long run relationships were analyzed using ARIMAX and ARDL models and future trends of financial inclusion were forecasted. All financial inclusion indicators have shown a significant improvement in the descriptive analysis of the study period. The average number of accounts was 106.26 million in 2023, up from 70.01 million in 2014, to 149.48 million in 2023. Likewise, the number of active mobile money accounts grew substantially from 9.50 million to 83.69 million, highlighting the swift growth of digital financial services in Bangladesh. The density of commercial bank branches and ATM coverage also showed positive growth trend during the study period. Mobile financial services have been identified as an important determinant of financial inclusion for the ARIMAX model. Forecasting performance of the model was acceptable and the test of the residuals was not rejected. According to the forecast, the number of account holders at banks will be further projected to rise during the forecast period and is expected to exceed 200.62 million by 2030. This implies that with current digital banking adoption trends, financial inclusion is likely to gain further in Bangladesh. The ARDL estimation results showed that there was a significant positive relationship between mobile

financial services and deposit accounts. The expansion of mobile banking has a positive and statistically significant coefficient, thus suggesting that mobile banking services help directly with the participation in the formal financial system. Results indicate that mobile financial services are an effective means to access previously unbanked and underbanked populations. The lagged account variable was determined to be positive and highly significant, meaning that there is high persistence in financial inclusion over time. About 74.5 percent of the current deposit accounts can be attributed to the previous year's level of deposit accounts, which reflects stability in the banking sector and continued growth in financial inclusion. The impact of commercial bank branches on financial inclusion was mixed. The contemporaneous effect of branch expansion was negative and the lagged effect was positive and significant. This discovery implies that expansion of branches can require short, term adjustment costs, but can have a positive impact on financial inclusion once the banking system is fully established. The Bounds Test gave evidence of a long run relationship between the number of deposit accounts, the number of mobile accounts, and the number of bank branches. The Error Correction Term (ECT) was negative and statistically significant, suggesting that the long run equilibrium is being corrected over the period. Based on the estimated adjustment speed, it is concluded that there exists a stable long run relationship between the variables, as around 28.3 % of disequilibrium is corrected within one year. The short, run Error Correction Model also confirmed that the adoption of mobile financial services is positively associated with increasing in deposit accounts with a strong positive coefficient. On the other hand, branch expansion had a negative direct impact in the short run, suggesting that digital financial services bring quicker gains in financial inclusion than traditional banking systems. The results of diagnostic tests using the model showed that the ARDL model was in a stable state and the residual of the ARDL model was normally distributed. But a serial correlation was observed in the residuals and the mobile account variable had a stationarity concern. Thus, caution needs to be exercised in interpreting the ARDL results and these results are used as additional evidence and not as the main source of forecasting. The findings of this study indicate that digital banking, particularly mobile financial services, plays a crucial role in promoting financial inclusion in Bangladesh. Though banking infrastructure is essential, mobile financial services have proven to be the best way to increase access to formal financial services. The forecasts further indicate that if the investment in digital financial infrastructure continues, financial inclusion will expand considerably and contribute to the sustainable economic development of Bangladesh.

Chapter 5

Conclusion and Recommendation

5.1 Conclusion

This study is aimed at analyzing financial inclusion by using the number and percentage of commercial bank deposit account holders in Bangladesh, digital infrastructure with active mobile money/MFS account holders, and conventional infrastructure with automated teller machines (ATMs) and commercial bank branches as proxy indicators, and study the relationship between these indicators of financial inclusion and digital and conventional infrastructures using ARIMAX and ARDL models and to predict the trend of financial inclusion in the short and long run from 2014 to 2023. The results for descriptive analysis showed significant growth from 2014 to 2023 in most indicators, including the average number of bank accounts (106.26 million), which increased from 70.01 million to 149.48 million; the number of active mobile money accounts (83.69 million), which rose significantly from 9.50 million to 83.69 million, reflecting rapid growth in digital financial services; and the density of commercial bank branches per 1,000 square km, which also grew positively. The effect of the share of MFS accounts on deposit accounts is positive and statistically significant both in the long run (ARDL model; $\beta = 0.1790$, $p = 0.0305$), and in the short run (Error Correction Model; ECM: $\beta = 0.2215$, $p = 0.0006$), with the latter having a stronger positive effect, suggesting that digital frameworks have more rapid impact on financial inclusion than traditional banking. The effect of commercial bank branches was also mixed; although the branch expansion effect in the short term was negative, though statistically insignificant ($\beta = -2.814$, $p = 0.0538$), but the effect of branch expansion in the long term was positive and statistically significant after about one, year of investment ($\beta = 3.0287$, $p = 0.0453$), it is that when a commercial bank invests in a new branch, it takes time to reach the maturity point, but after that period it has a positive effect. The previous year's level of the dependent account variable, the lagged dependent account variable, was found to be positive and highly significant ($r = 0.7454$, $p = 0.0026$), indicating a high degree of persistence and stability over time; that is, 74.5% to 75% of the dependent account variable was explained by the previous year's level. The statistical results were inconclusive, but the results showed that ATMs have positive impacts, as indicated by positive coefficients in both the ARIMAX and ECM models, however, only the coefficient of ECM is statistically significant ($\beta = 839.65$, $p = 0.0002$), meaning ATMs are less impactful on financial

inclusion compared to mobile financial services. The ARDL Bounds Test obtained an F, statistic of 4.1476 that is above the lower critical bound (5%) and upper critical bound (10%) supporting the existence of a stable, long, run equilibrium relationship between deposit account, mobile account and bank branch, albeit the results were inconclusive at a strict 5% level. The Error Correction Term ($ECT = 0.2832$, $p = 0.0012$) is negative and statistically significant, indicating that the short, run disequilibrium will be corrected by 28.3% within one year and gradually move back to the equilibrium path following a shock. The ARDL model was found to be mixed in terms of model adequacy as the model was in a stable state and the residuals were also close to normality as indicated by the Jarque–Bera test ($p = 0.6476$), but the Breusch–Godfrey test found a remaining residual serial correlation ($p = 0.0027$), and finally stationarity results suggested that the data contained was limited as the deposit and branch variables were stationary after first differencing ($I(1)$), while the mobile services variable remained non, stationary even after differencing. Although this mixed order of integration is not a problem for the ARDL model, since it does not require two variables to be integrated at a time, the non, stationarity of the mobile variable requires some caution and further research needs to be conducted to validate these findings with other unit root tests conducted using longer time, series data, and these ARDL results are used as additional corroborative evidence in the forecasting process and not as the primary evidence. Having acceptable performance, good fit of the residuals to the historical data, and no statistically significant residual autocorrelation (Ljung-Box test, $p = 0.1436$), an ARIMA(0,1,0) process with exogenous variables (ARIMAX) was decided to be used to show the historical trend and produce reliable forecasts. Bank deposit accounts are expected to increase steadily from 155.63 million in 2024 to surpass 200.62 million by 2030, with the annual growth rate in financial inclusion expected to gradually decline to 3.88% in 2030, as the market matures over the years. Ultimately, these findings indicate that digital banking, particularly mobile financial services, plays a crucial role in promoting financial inclusion in Bangladesh, proving to be the best way to increase access to formal finance, and implying that if investments in digital financial infrastructure continue, financial inclusion will expand considerably and contribute to sustainable economic development.

5.2 Recommendations

This study recommends the following measures: Based on the empirical results of this study, several recommendations are put forward to enhance the financial inclusion in Bangladesh through online banking and digital financial services: The ARDL results revealed that active mobile financial service (MFS) accounts had the statistically significant positive effect both in the long run ($\beta = 0.1790$, $p = 0.0305$) and short run ($\beta = 0.2215$, $p = 0.0006$) periods, making it the best explanation of financial inclusion. Accordingly, it is critical for policy makers to focus on the following areas to enhance mobile financial services: grow agent networks, strengthen interoperability between banks and mobile financial service providers, and bring digital financial services to remote and under-served communities. These are to be expected to spur unbanked populations to formal financial sector.

The results also show that the impact of commercial bank branches on financial inclusion is found as a delayed effect. The contemporaneous branch coefficient was negative ($\beta = -2.8145$) while the lagged coefficient became positive and statistically significant ($\beta = 3.0287$, $p = 0.0453$), suggesting that improvements in physical banking assets take time to reap tangible rewards. It is therefore important that commercial banks continue to grow the number of their branches in strategic locations especially in rural and economically disadvantaged areas, and note that returns from such investments will be likely to be realised over time as opposed to all at once.

ATM expansion showed a positive association with financial inclusion but statistically did not impact. This implies that expanding ATM services does not necessarily lead to a significant boost in financial inclusion. In addition to promotion of the ATM network, it is essential for the financial system to strengthen the efficiency, access and functionality of the existing ATM network by linking existing ATMs with comprehensive digital banking platforms and deliver continuous services.

The estimated value of the lagged dependent variable ($\beta = 0.7454$, $p = 0.0026$) suggests that there is a high level of persistence in the growth of deposit accounts, and thus, improvements in financial inclusion tend to be persistent over time. Furthermore, the high magnitude of the Error Correction Term ($ECT = -0.2832$, $p = 0.0012$) is indicative of a fairly stable LR adjustment process. This indicates that policy consistency in financial inclusion

should be kept and that policies should not be changed too often as this could hinder the full impact of digital financial reforms in the long, run.

The number of commercial bank deposit accounts are projected to grow steadily from around 155.63 million in 2024 to 200.62 million by 2030, with a deceleration in the annual growth rate from 4.82% to 3.88% during the forecast period. The projections suggest that financial inclusion in Bangladesh is likely to grow even further but with a possible slowdown when markets become more saturated. Hence, Bangladesh Bank and commercial banks need to leverage such predictions in the implementation of their long, term financial inclusion strategies, resource allocations, and investment in financial infrastructure to meet the increasing demands of users of financial services through digital channels.

According to the diagnostic tests the ARIMAX model is an acceptable model for the data, with the Ljung–Box test showing there is no significant residual autocorrelation. The Breusch, Godfrey test revealed serial correlation in the ECM residuals while the unit root test revealed persistent non, stationarity in the mobile financial service variable. Such constraints did not preclude meaningful estimation and forecasting, but in the future longer time, series data sets, more explanatory variables, other unit root tests, and more sophisticated econometric methods are recommended to enhance the robustness of the models and their forecasting power.

Any effort for sustainable financial inclusion must entail coordinated action by Bangladesh Bank and commercial banks, mobile financial service providers, the telecom sector, and government institutions. In addition to the expansion of digital financial services, financial literacy should be improved, cybersecurity should be strengthened, consumers protection should be enhanced, and digital trust should be fostered among the consumers. With this integration of policy measures, the further growth of online banking could be expected to contribute to more extensive and inclusive and sustainable financial development across Bangladesh.

5.3 Recommendations for Future Study

- This research has brought significant evidence about the role of online banking in financial inclusion in Bangladesh but there are some opportunities for future studies. First, the study relied on secondary data obtained on an annual basis over a period of 11 years (from 2014 to 2023). Longer time, series data and higher, frequency data

(quarterly, monthly) would be beneficial in future studies to increase the accuracy of the estimate and forecasting performance.

- Secondly, this study looked at the type of financial services as the main instrument that was used in determining financial inclusion, and explanatory variables were active mobile financial services accounts and ATMs and commercial bank branches. Future studies could further include other variables like internet banking users, digital payment transactions, mobile banking transaction value, debit card users, credit card users, agent banking activities, and number of loan accounts, as well as fintech adoption to have a proper and a thorough evaluation of financial inclusion.
- Third, the ARDL model and ARIMAX model were used in this study to obtain meaningful results, but in the future, other econometric techniques are used instead, such as Vector Error Correction Models (VECM), Vector Autoregression (VAR), Nonlinear ARDL (NARDL), Bayesian time series modeling, and machine learning forecasting techniques, to compare the results and get more robust estimates.
- Fourthly, the current research studied financial inclusion at the national level on macroeconomic data. Further work could focus on regional differences, urban and rural differences, financial inclusion for women, or on the household level on the adoption of online banking services. These analyses would offer more insights into how various population groups are faring with regard to digital financial inclusion.
- Lastly, this research found some methodological drawbacks such as the constant non-stationarity of the mobile financial service variable, and serial correlation of the Error Correction Model's residuals. For future research, the use of alternative unit root tests, with larger sample sizes and other explanatory variables can be used to enhance the model specification and the reliability of empirical results.

5.4 Limitations

- The study is based on limited time-series observations with 10 years data (2014-2023) of Bangladesh Bank from secondary sources. The relatively small sample size may lead to a decrease in statistical power of the econometric analysis and reduce the accuracy of long-run estimates.
- The second limitation of the explanatory variables used for financial inclusion is that it was only explained by financial instruments that are currently in use (ATMs, commercial banking and active mobile financial service (MFS) accounts). Some other

factors like Internet banking users, digital payments, debit card, credit card, agent banking, use of fintech and income level, were excluded due to data availability.

- The study treats the commercial bank deposit account as proxy measure of financial inclusion. While deposit accounts are a common metric, they do not necessarily reflect all elements of financial inclusion, including access, use, cost, financial services quality, and financial literacy.
- The mobile financial service variable was also found to be non-stationary by the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) tests. This is a limitation that must be considered when interpreting the results related to this variable, although the model resulted in meaningful estimates and forecasts.
- The Breusch–Godfrey Serial Correlation Test indicated that there was autocorrelation in the ECM residuals, which was evidence of one of the classical regression assumptions not being met. The coefficients estimated were efficient, but the model was stable and gave a good empirical evidence, but serial correlation might have an effect on the efficiency of the estimated coefficients.
- National level analysis: The analysis has been done on the national level, which does not reflect the differences in financial inclusion and online banking adoption at the regional, urban–rural, gender, and household level in Bangladesh.
- The ARIMAX model can be used to forecast deposit accounts growth up to 2030 assuming that there are historical trends and relationships between the variables. The actual future performance may be affected by unexpected changes of economic conditions, regulatory measures, new technologies or changes in the structure of the financial services industry, and such changes could affect the accuracy of the forecasts.
- Secondary data dependence: The study is purely based on secondary data obtained from Bangladesh Bank. The empirical results may be influenced by any errors, revisions or omissions in the official data sources.

5.5 Declaration of generative AI use

During the preparation of this work, ChatGPT was used for basic grammar correction.

Chapter Six

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